Final Project

NTSC Video Processing System

ENSC 425 – Electronic System Design

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By: Kevin Shen

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Introduction

The purpose of this project was to design and implement a circuit capable of decoding a standard NTSC composite video signal and displaying the resulting image on the cathode ray tube (CRT) of an analog oscilloscope, using the X-, Y-, and Z-axis inputs for horizontal deflection, vertical deflection, and beam-intensity control, respectively. The system separates the horizontal and vertical synchronization signals to generate the required sweep waveforms while processing the luminance signal separately for image display. The overall design consists of input processing, sync separation, sweep generation, blanking, and luminance processing subcircuits, which were individually designed, tested, and then integrated into the final system.

High-Level Specification

Input Signal:
Standard NTSC video, RCA connector with adapter.

Three Output Signals:
$V_{H\text{-ramp}}$: Applied to X-Channel of oscilloscope.
$V_{V\text{-ramp}}$: Applied to Y-Channel of oscilloscope.
V_{Lumin}: Applied to Z-Channel of oscilloscope.

Controls:
Two trim-pots to adjust Contrast and Brightness.

Timing:
The video timing should correctly display odd/even fields with accurate interlace timing.

Blanking:
The luminance output signal should be blanked during both the horizontal and vertical porches.

Overscan:
When the input signal is not present, both ramp signals should be free-running and continue to scan. Obviously sync extraction will not be possible so a reasonable amount of overscan should be specified.

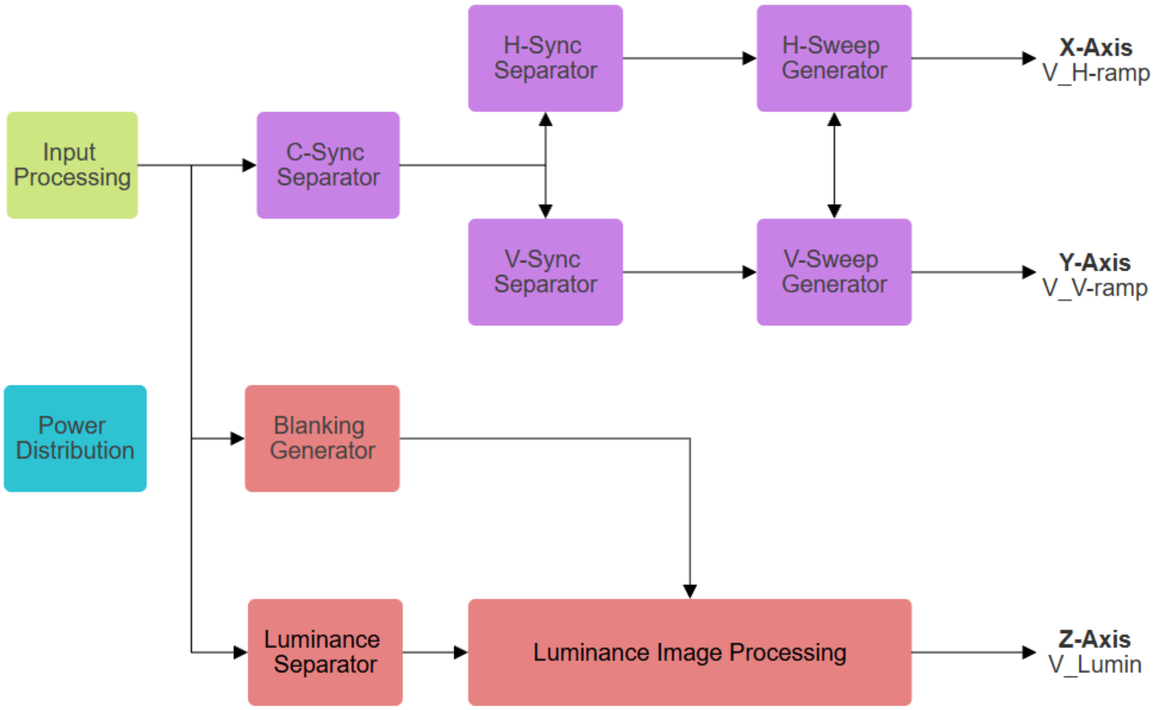

Figure 1: High-Level Block Diagram of the NTSC Video Processing System

Subcircuit Interface Specifications

Power Supply Levels

Function

- Provide the required supply voltages for each circuit section.
- Generate a 5 V supply from the available ±12 V supplies.

Requirements

- Provide stable +12 V and −12 V supply rails.
- Provide a regulated +5 V supply for ICs requiring lower supply voltages.
- Keep all supply voltages within component ratings.

Inputs

- +12 V and −12 V laboratory power supplies.

Outputs

- +12 V and −12 V rails.
- +5 V supply rail.

Input Buffering, Clamping
Function:
- Isolates the input signal and produces a stable signal for further processing.

Requirements:
- Output should be independent of average frame luminance.
- Preserve the required video bandwidth with minimal distortion.
- Provide sufficient isolation from downstream loading.

Inputs:
- Standard NTSC composite video signal.

Outputs:
- Buffered and clamped NTSC composite video signal.

Composite Sync Separator
Function:
- Extract only the synchronization information from the NTSC signal.

Requirements:
- All sync tips must be present with minimal delay and reasonably sharp edges.

Inputs:
- Buffered and clamped NTSC composite video signal.

Outputs:
- Composite sync signal containing the NTSC synchronization pulses.

Horizontal Sync Separator
Function:
- Produce horizontal sync pulses only.

Requirements:
- Horizontal sync pulses should occur only at the line frequency.
- Output pulses should be approximately 4 μs, which are typical for H-sync tips. Equalization pulses during the vertical retrace period must be correctly processed, producing a continuous stream of H-sync pulses at the line frequency.

Inputs:
- Composite sync signal.

Outputs:
- Horizontal sync pulse signal.

Vertical Sync Separator

Function:

- Processes the equalization pulses during the vertical retract period to produce the V-Sync pulses.

Requirements:

- V-sync pulses should have reasonably sharp edges.
- Latency should be no more than 25% of one horizontal line period.

Inputs:

- Composite sync signal.

Outputs:

- Vertical sync pulse signal.

Horizontal and Vertical Sweep Generators

Function:

- Produce free-running horizontal and vertical ramp signals for raster scanning.

Requirements:

- Ramps should be sufficiently linear.
- Ramps must continue running when the NTSC input or sync pulses are absent.
- H- and V-ramp reset times should be short relative to their respective retrace periods.

Inputs:

- H-sync pulse signal for the horizontal sweep generator.
- V-sync pulse signal for the vertical sweep generator.

Outputs:

- V_{H-ramp} to the oscilloscope X-axis.
- V_{V-ramp} to the oscilloscope Y-axis.

Blanking Generator

Function:

- Produce a blanking flag during the horizontal and vertical retrace periods.
- Keep the flag active through the front porch, sync tip, and back porch.

Requirements:

- Blanking must cover the full horizontal and vertical retrace periods.
- Latency should be small relative to the horizontal retrace period.

Inputs:

- Horizontal sync signal.
- Vertical sync signal.

Outputs:
- Blanking flag signal for the luminance processing stage.

Luminance Separator
Function:
- Removes the synchronization information, leaving only the luminance information.

Requirements:
- Maintain sufficient bandwidth to preserve the luminance/image information.
- Minimize distortion of the luminance signal.

Inputs:
- Buffered and clamped NTSC composite video signal.

Outputs:
- Intermediate luminance signal with sync information removed.

Luminance Image Processing
Function:
- Combines the intermediate luminance signal with the blanking flag and allows adjustment of the brightness and contrast of the final image.

Requirements:
- Maximum and minimum output levels must match the oscilloscope Z-axis requirements.
- The black level during the blanking periods should be specified in terms of the chosen maximum/minimum levels of the luminance.
- Adjustment of the contrast and brightness should not change the black level.

Inputs:
- Intermediate luminance signal.
- Blanking flag.

Outputs:
- Final luminance signal V_{Lumin} for the oscilloscope Z-axis.

Design Activity

Power Supply Levels

Initial power distribution utilized +5 V and +12 V supply channels to power the remaining logic and the vertical/horizontal ramp generators, respectively. Integrating the luminance inverting amplifier required a negative rail, prompting a transition to a dual ± 12 V supply. To maintain 5 V logic levels for the LM311 comparators, +5 V was derived from the +12 V rail using a potentiometer voltage divider. Because direct current drawn by peripheral ICs (555 timer, LM311) introduced loading effects and voltage fluctuations on this derived rail, an LM358 unity-gain buffer was inserted to isolate the divider and provide a stable, low-impedance +5 V supply.

Input Buffering, Clamping

A DC clamp is required for the NTSC signal to restore a fixed reference baseline lost to variations in the average video voltage. Due to hardware constraints, specifically the availability of only a single high-speed op-amp, a passive clamp was selected over an active clamp design. This allocation preserved the high-speed op-amp for the luminance amplification stage. Because an active clamp using a lower slew-rate op-amp would have severely degraded high-frequency video detail, relying on a passive architecture represented the primary design trade-off implemented to maintain overall signal integrity.

The clamps reference voltage is calculated by:

$$V_{ref} = 12 \cdot \frac{2.2k}{2.2k + 39k} = 0.641\, V$$

Where V_D is was measured to be 0.62 V

$$V_{clamp} = V_{ref} - V_D = 0.641 - 0.62 = 0.021\, V$$

Thus these resistor values for the voltage divider were chosen so the sync tips could be held slightly above 0V.

The color burst signal is suppressed because chrominance processing is omitted from this architecture, establishing a more accurate reference voltage for the composite sync separator comparator. Due to the component constraints noted previously, a passive low-pass filter is implemented rather than an active topology

Since the NTSC color burst subcarrier is centered at 3.58 MHz, the filter attenuates its amplitude without distorting critical baseband video components. A cutoff frequency of 1.59 MHz was selected because the luminance signal energy is heavily concentrated on the lower end of the 4.2 MHz NTSC bandwidth.

$$|H(f)| = \frac{1}{\sqrt{1+\left(\frac{f}{f_c}\right)^2}} = \frac{1}{\sqrt{1+\left(\frac{3.58M}{1.59M}\right)^2}} = 0.4059$$

A cutoff frequency of 1.59 MHz gives us an amplitude reduction of the colour burst of around 59.4% (1 - 0.4059). This cutoff frequency effectively dampens the high-frequency colour burst while preserving fundamental luminance data.

Although having a video amplifier at the input stage is theoretically optimal for signal buffering and gain control, the raw NTSC input amplitude provided sufficient dynamic range for accurate measurement and sync extraction. Due to physical component constraints and to preserve subcircuit independence, input amplification was omitted in the physical implementation.

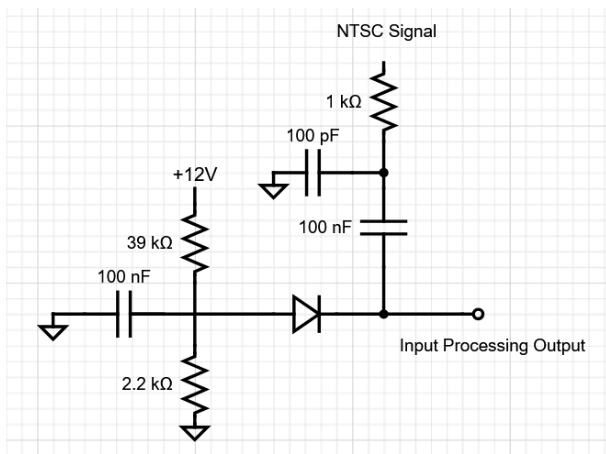

Figure 2: Input Clamping Circuit

Limitations of Passive Clamp with live NTSC Video

As shown in the oscilloscope capture, Channel 2 (cyan) displays the clamped waveform using the diode clamp circuit, while Channel 1 (yellow) represents the synthesized NTSC input signal. Although the passive clamp operates effectively with the controlled artificial NTSC signal, testing with a live NTSC source revealed stability issues. Due to rapid variations in Average Picture Level and dynamic video content in real-world signals, the passive clamp's time constant prevents it from maintaining a constant DC baseline, causing the clamped level to drift or swing.

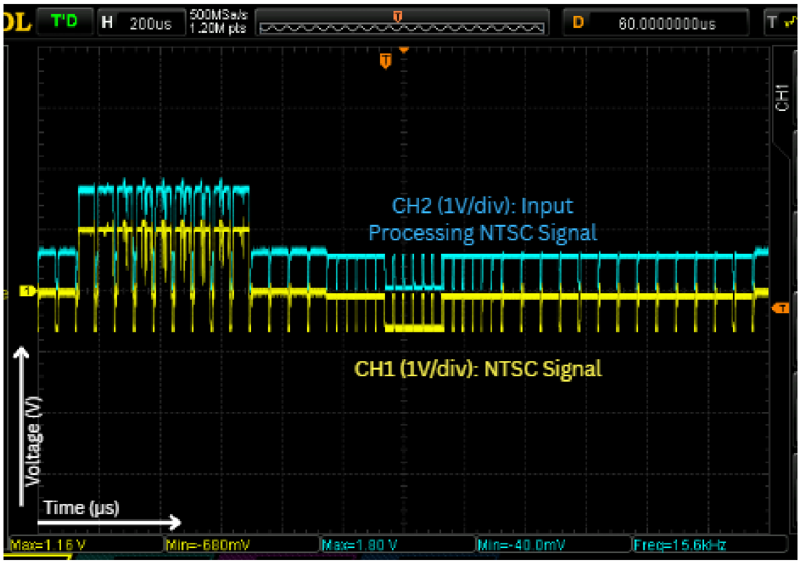

Figure 3: NTSC Passive Clamp Capture

Passive Clamp Loading Anomaly and System Integration

When integrating the passive clamp with the subsequent composite sync (C-sync) separator stage, an unexpected loading anomaly occurred: the C-sync separator failed to operate unless an oscilloscope probe was actively connected to the clamp's output node. This behavior suggested that the clamp output was presenting an unstable, high-impedance node to the comparator, lacking an adequate DC discharge path and allowing parasitic noise to disrupt proper triggering. The 10MΩ input resistance and parallel capacitance of the oscilloscope probe provided the necessary bleed path and high-frequency filtering required to stabilize the node. To resolve this issue in the hardware, a 1MΩ resistor was added from the clamp output to ground, providing a permanent discharge path that restored proper C-sync separator operation without requiring external probing.

Composite Sync Separator

First Iteration

The composite sync separator utilizes an LM311 comparator, selected for its low propagation delay and single-supply capability. Because the LM311 features an open-collector output, a pull-up resistor is required to establish valid logic levels. An active-low output configuration ensures the signal remains HIGH by default and drops LOW upon detecting a sync pulse. Consequently, the inverting terminal receives the reference voltage, while the non-inverting terminal receives the signal from the input buffering and clamping stage.

From measuring from the sync tip (4 mV) to right below the colour burst (348 mV)

$$V_{ref} = \frac{348 + 4}{2} = 176\,mV$$

Calculating the voltage divider

$$176mV = 5 \cdot \frac{R_2}{R_2 + R_1}$$

$$R_1 = 27.41\,R_2$$

Choosing $R_2 = 1k\Omega$, $R_1 = 27k\Omega$

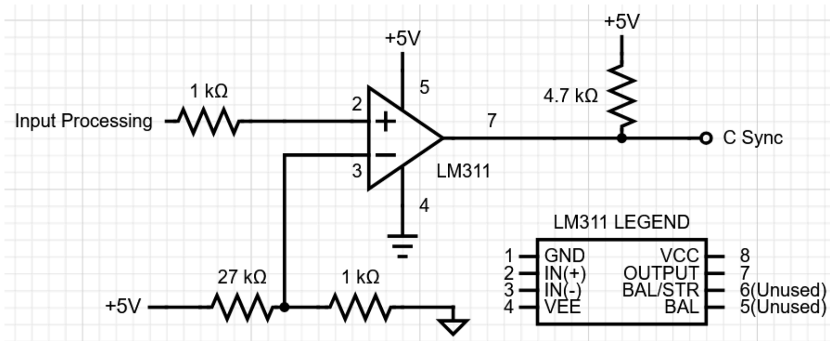

Figure 4: First Iteration C-Sync Separator Circuit

Second Iteration

While the initial iteration performed reliably when connected solely to the input buffering and clamping stage, interfacing it with the horizontal sync separator induced instability in the composite sync output. Datasheet analysis for the LM311 revealed that the initial pull-up resistance was excessively high, necessitating a replacement with a 220Ω resistor. Additionally, a 1MΩ positive feedback resistor was introduced to establish hysteresis, suppressing small noise spikes and preventing rapid output oscillation. These modifications successfully restored stability to the output signal .

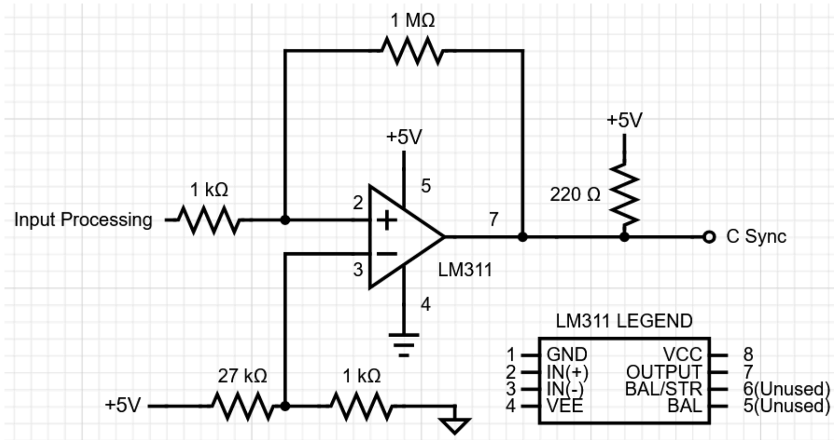

Figure 5: Second Iteration C-Sync Separator Circuit

Vertical Sync Separator

First iteration

Since the vertical sync pulse has a frequency of 59.94 Hz, a low pass filter with a cutoff frequency of 1591 Hz was selected to isolate the vertical sync pulses. A passive low-pass filter was determined to be preferable by prioritizing circuit simplicity and stability over active gain. By using only a resistor and capacitor, it eliminates additional op-amps and active feedback stages, reducing board space and power consumption.

$$1591 = \frac{1}{2\pi RC}$$

$$RC = \frac{1}{9996.54}$$

Choosing R = 10kΩ, C = 10nF

Although the passive filter yields gentler roll-off slopes, this is negligible in this application because the integrated output is directly fed into a second stage utilizing a 555 timer with the inputs pins THRESH and TRIG connected to create hysteresis to prevent chattering and restore sharp digital transitions. A 10 nF decoupling capacitor was added to pin 5 to ground to filter out noise.

Second Iteration

Although a 1,591 Hz cutoff frequency is theoretically ideal for isolating vertical sync pulses, testing revealed that the heavily attenuated output spikes were insufficient to trigger the 555 timer. Replacing the 10 nF capacitor with a 1 nF capacitor increased the cutoff frequency to 15.9 kHz, significantly boosting the pulse amplitude for reliable 555 triggering. While this higher cutoff frequency allowed 15.734 kHz horizontal serration ripples to pass through, these high-frequency dips were effectively rejected by the secondary 555 stage due to its built-in hysteresis thresholds, preserving clean vertical sync generation.

Horizontal Sync Separator

First Iteration

For the horizontal sync separator, since the horizontal sync pulse has a frequency of 15.734 kHz, the initial design used a similar approach to the vertical sync separator but instead using a passive high pass filter with a cutoff frequency of 15.915 kHz as the first stage then into a second stage using a 555 timer with hysteresis to produce the h-sync separator output.

$$15915 = \frac{1}{2\pi RC}$$
$$RC = \frac{1}{99996}$$

Choosing R = 1kΩ, C = 10nF

Although this method worked for the vertical sync separator, it ultimately failed for the horizontal sync separator because only the 15.734 kHz frequency of the h-sync pulse needs to be captured, but during the vertical blanking interval the NTSC introduces equalizing pulse and vertical serration at twice the horizontal sync pulse frequency 31.468 kHz.

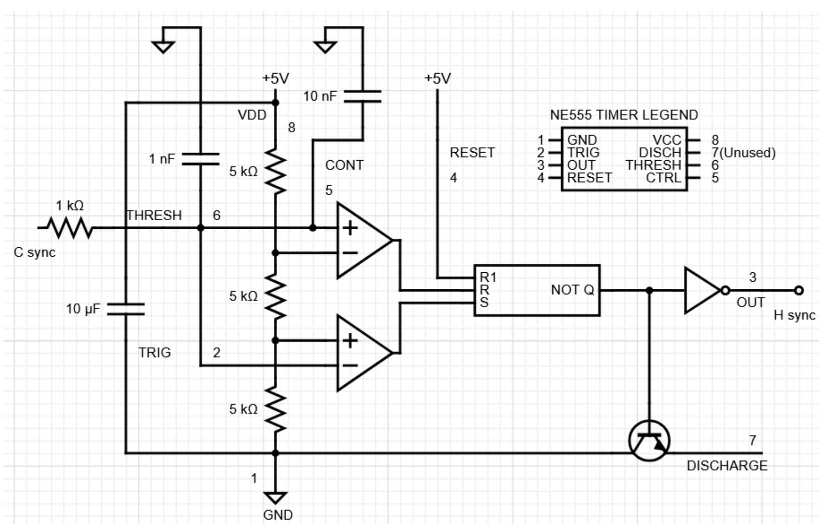

Figure 6: First Iteration H-Sync Circuit

Second Iteration

For the second approach, a monostable multivibrator constructed from 555 timers was used. To configure a 555 timer as a monostable multivibrator, the DISCHARGE pin and THRESHOLD pin are connected together to a node between a capacitor and resistor to charge the capacitor. Therefore, the RC charging formula is used to determine the width of the pulse.

$$V(t) = V_{CC} - (V_{CC} - V_{intial})e^{-\frac{t}{RC}}$$

Since the DISCHARGE and THRESHOLD pins are connected, to calculate the charge time from 0V to ⅔ V_{CC} . Therefore, V(t) = ⅔ V_{CC} , $V_{intial} = 0$.

$$\frac{2}{3}V_{CC} = V_{CC} - V_{CC}e^{-\frac{t}{RC}}$$

Solving for t gives us

$$t = ln(3) \cdot RC = 1.0986 \cdot RC$$

By setting its output pulse duration to approximately 45 µs, a monostable multivibrator creates a blind window that naturally filters out double-frequency pulses while passing normal horizontal sync. Standard horizontal lines arrive every 63.5 µs (15.7 kHz), so when a line edge triggers the timer at t = 0 µs, the output stays active for 45 µs, finishes, and resets in time to catch the next line pulse at 63.5 µs. However, during vertical blanking, equalizing and serration pulses arrive at twice the rate at every 31.75 µs (31.5 kHz). When the first edge fires the timer, the double-frequency pulse arrives at 31.75 µs while the monostable is still in its active 45 µs period, causing the circuit to completely ignore the extra edge. Because 45 µs sits cleanly between 31.75 µs and 63.5 µs, the timer masks out every intermediate pulse and preserves a continuous 15.7 kHz output rate across the entire frame.

$$\tau = 1.1(3.9k)(10n) = 42.9\ \mu s$$

Choosing R = 3.9kΩ, C = 10nF

But this design produced an unstable output and unwanted feedback into the previous subcircuit stage, introducing unwanted noise. To resolve this instability, an LM311 comparator stage was used between the composite sync output and the H-sync separator input to serve as a digital buffer. Using an equal-resistor voltage divider, the LM311 reference voltage was set to 2.5 V, the midpoint of the 0 V to 5 V C-sync signal, ensuring noise rejection and stage isolation. Although this resolved the feedback problem, it did not fix the unstable output.

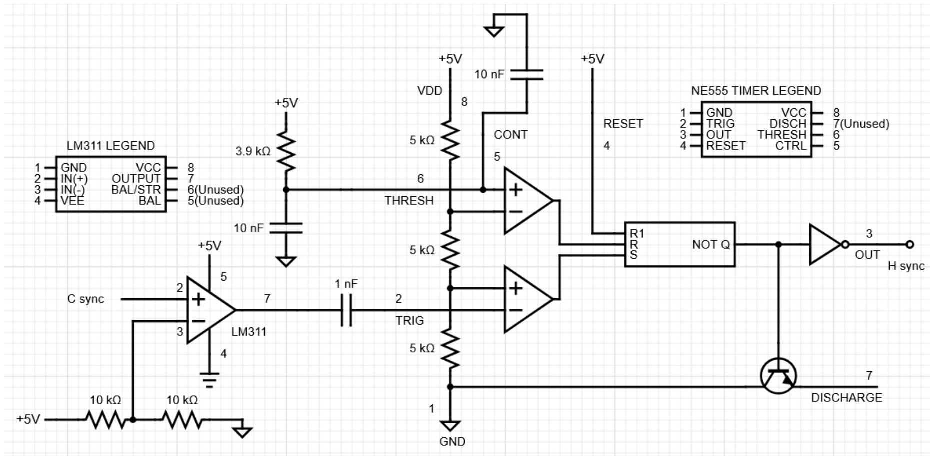

Figure 7: Second Iteration H-Sync Circuit

Third Iteration

In this third design, the instability issue is addressed by using a dual-stage masking approach using monostable multivibrators. Dual-stage masking fixes the instability issue by dividing the workload between two interlocked timers: one dedicated exclusively to establishing a time-masking window and the other to generating the clean output pulse. In a single 555 timer monostable, attempting to mask out double-frequency pulses while simultaneously triggering off raw, level-sensitive sync edges causes trigger lockup and output jitter. The dual-stage setup solves this because Timer 1 fires first to create a 42.9 µs masking window that activates an external NPN transistor, which physically locks out the second timer and creates a dead zone during which any 31.468 kHz double-frequency pulses at 31.75 µs are completely blocked. By actively clamping Timer 2 during this window, the circuit prevents overlapping trigger conflicts, allows the timing capacitors to fully discharge back to 0 V, and ensures Timer 2 only fires on valid 15.7 kHz line edges, eliminating output chattering, jitter, and pulse-width drift across the vertical blanking interval.

Timer 1 pulse width:

$$t = 1.1(39k)(1n) = 42.9\ \mu s$$

Timer 2 pulse width:

$$t = 1.1(47k)(1n) = 51.7\ \mu s$$

To ensure reliable functionality an AC-coupling network consisting of a 1 nF series capacitor, a 10 kΩ pull-up resistor, and a clamping diode was implemented at each trigger pin to prevent trigger lockup by turning raw sync edges into short spikes, while clamping positive voltage overshoot to protect the chip from ringing and false re-triggering.

While the application relies strictly on the falling edges of the pulses for line timing, the trade-off of generating a 51.7 µs output pulse is that it consumes nearly majority of the 63.55 µs horizontal line period, drastically shortening the timing capacitor's recovery window to just 11.8 µs and stripping away the standard video line structure. Furthermore, because this extended pulse completely eliminates the front and back porches, it breaks compatibility with circuits that depend on a standard 4.7 µs pulse width and a dedicated back-porch sampling interval.

The full third iteration H-sync circuit diagram can be found in the appendix Figure 35.

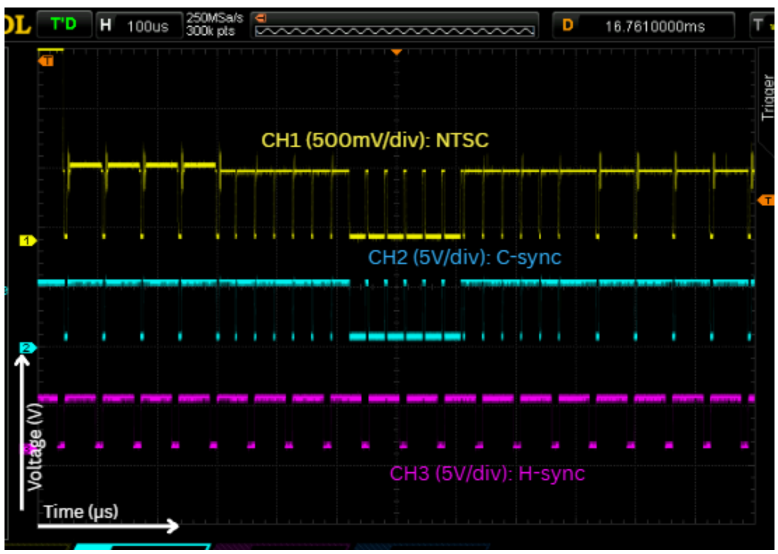

Figure 8: C-sync and H-sync Oscilloscope Captures

Fourth Iteration

As shown in the third iteration, the generated pulse width measured 51.7 µs due to a component selection error, as a 47 kΩ resistor was mistakenly installed instead of the intended 4.7 kΩ value. Correcting the resistor value reduces the RC time constant by a factor of ten, yielding the expected 5.17 µs pulse width. You can see the changes in the screen capture below.

Timer 2 pulse width:

$$t = 1.1(4.7k)(1n) = 5.17\ \mu s$$

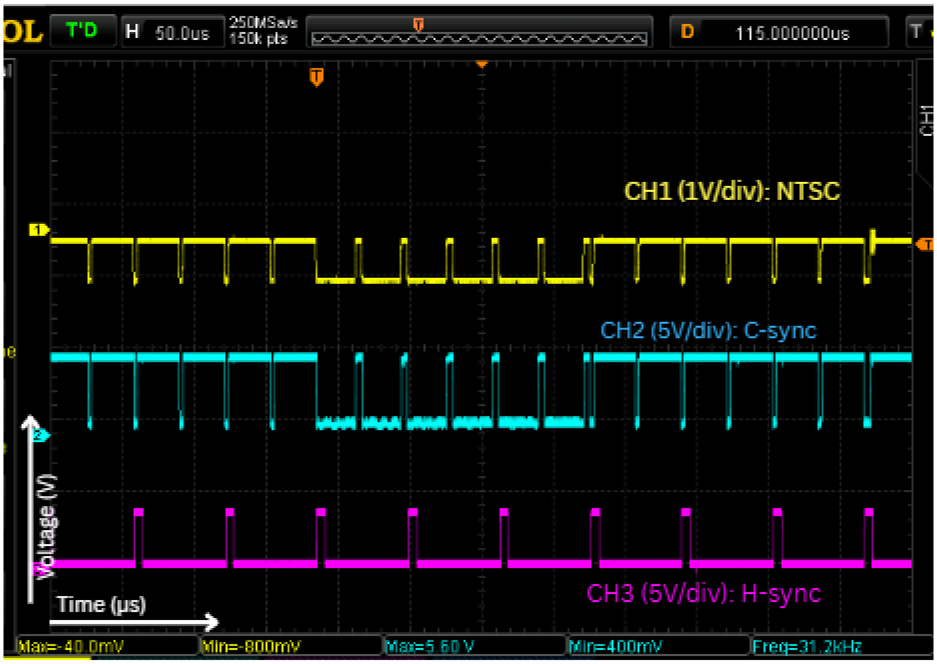

Figure 9: Fourth Iteration H-sync Oscilloscope Capture

Horizontal and Vertical Sweep Generator

First Iteration

For the horizontal and vertical sweep generator, 555 timers were used to take advantage of the internal discharge switch and threshold comparators to create the ramps. Ramp generators can be created by measuring the voltage of a capacitor as current from a resistor charge it, and by connecting them to the DISCHARGE pin, the voltage in the capacitor is allowed to discharge to ground through this pin, and by connecting the DISCHARGE pin to the TRIG pin this allows the capacitor to recharge whenever the capacitor dissipates to ⅓ Vcc. But since a resistor is used to charge up the capacitor, the voltage in the capacitor does not charge up linearly. So a smaller section of the voltage curve is used so it will seem more linear. Thus, the output of the V-sync or H-sync separators is connected into the THRESHOLD pin of the 555 timer. An AC coupling capacitor is used at the output using a 100 nF capacitor and at the input using a 10 nF at pin 5.

Since the vertical sync pulses have a frequency of 59.94 Hz, the capacitor is designed to charge over the 16.68 ms field period to 1V above ⅓ Vcc in 16.7 ms (1/59.94). To calculate the component values, the RC charging formula is applied.

$$V(t) = V_{CC} - (V_{CC} - V_{intial})e^{-\frac{t}{RC}}$$

To start charging from ⅓ V_{CC} to 1 + ⅓ V_{CC}. Then V_{intial} = ⅓ V_{CC} , V(t) = 1 + ⅓V_{CC} Substituting the desired voltage ranges.

$$1 + \frac{V_{CC}}{3} = V_{CC} - (V_{CC} - \frac{V_{CC}}{3})e^{-\frac{t}{RC}}$$

Solving for t give us

$$t = RC \cdot ln\left(\frac{2V_{CC}}{2V_{CC} - 3}\right)$$

Since t = 16.7 ms, V_{CC} = 5V choose C = 10µF

$$R = \frac{\tau}{C \cdot ln\left(\frac{2 \cdot V_{CC}}{2 \cdot V_{CC} - 3}\right)} = \frac{16.7m}{(10\mu) ln\left(\frac{10}{10-3}\right)} = 4682\ \Omega$$

Thus choose R = 4.7kΩ

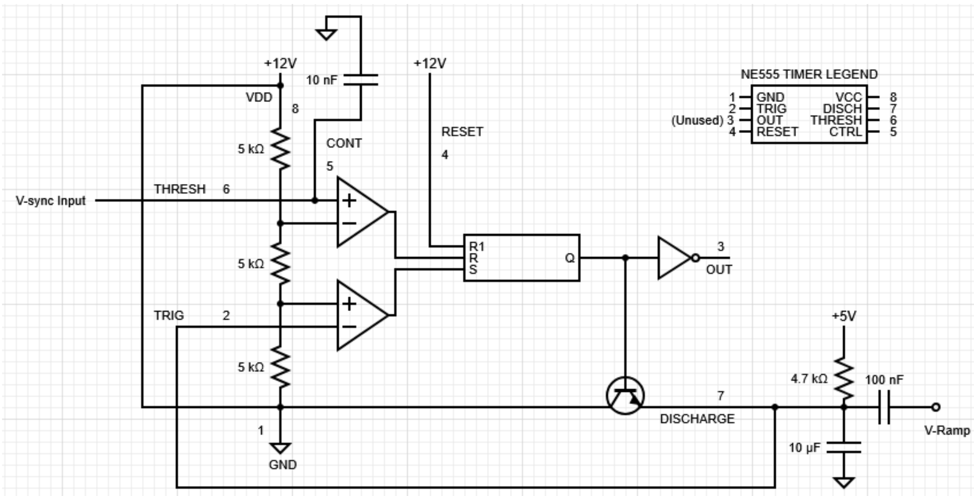

Figure 10: First Iteration H-Ramp/V-Ramp Circuit

Second Iteration

Although the first iteration subcircuit provided sufficient ramps with slight but noticeable curvature in the slope, the linearity of the slope can be greatly improved just by replacing the resistor with a Wilson current mirror, and if the voltage amplitude of the ramps is increased, the noise relative to the ramp will look smaller. Since the supply voltage is increased, the threshold level will also increase, thus a DC offset is required at the input to pull the voltage up.
Since the supply voltage was increased to 12V, the threshold level increased to 8V (⅔ VCC). A voltage divider is used to bring up the idle voltage.

$$V = 12 \cdot \frac{15k}{15k+10k} = 7.2V$$

Here, 15kΩ and 10kΩ resistors are used to bring up the idle voltage to 7.2 V, right below the threshold level. The capacitor is designed to charge up to 6V, recalculating the capacitor and resistor value using the formula for charging a capacitor with a constant current source.

$$t = \frac{C \cdot \Delta V}{I}$$

Using t = 16.7ms, ΔV = 6, choosing C = 10μF

$$I = \frac{C \cdot \Delta V}{t} = \frac{(10\mu)(6)}{16.7m} = 3.59mA$$

Calculating resistance needed for wilson current mirror

$$R = \frac{12 - 0.7 - 0.7}{3.59m} = 2952\Omega$$

Thus choose R = 3.3kΩ

Since H sync pulses have a frequency of 15.734 kHz, the capacitor is designed to charge up 6V in t = 63.5μs (1/15.734k). Choosing C = 47nF.

$$I = \frac{C \cdot \Delta V}{t} = \frac{(47n)(6)}{63.5\mu} = 4.44\ mA$$

Calculating resistance needed for the Wilson current mirror:

$$R = \frac{12 - 0.7 - 0.7}{4.44m} = 2387\ \Omega$$

Thus choose R = 2.2kΩ.

The full second iteration H-ramp and V-ramp circuit diagram can be found in the appendix Figure 36.

H-sync Separator Fix

Although no direct modifications were made to the horizontal ramp generator, resolving the issues within the horizontal sync separator restored proper ramp resetting. Fixing the horizontal sync separator ensured that reliable reset pulses reached the ramp generator, eliminating the failure where the ramp would not discharge, as demonstrated in the before and after waveforms below.

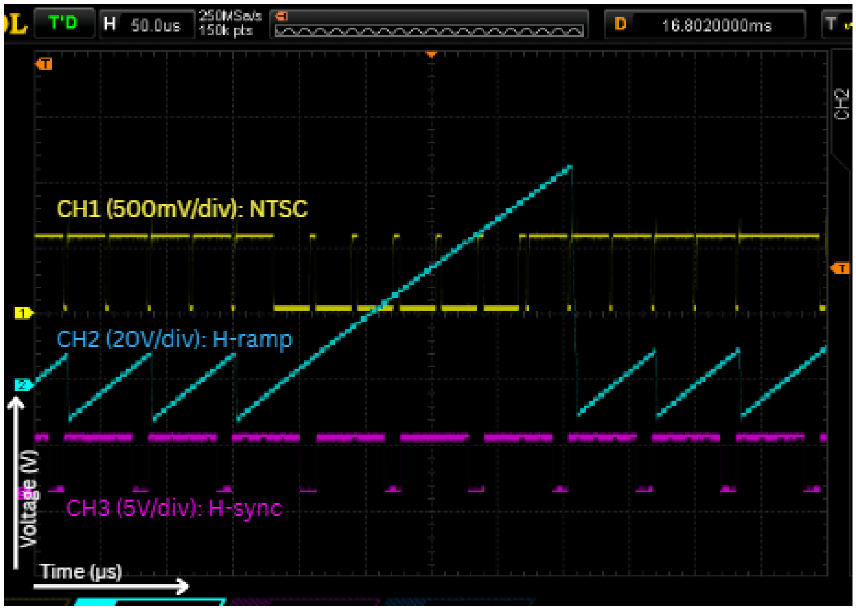

Figure 11: Before H-sync Separator Change

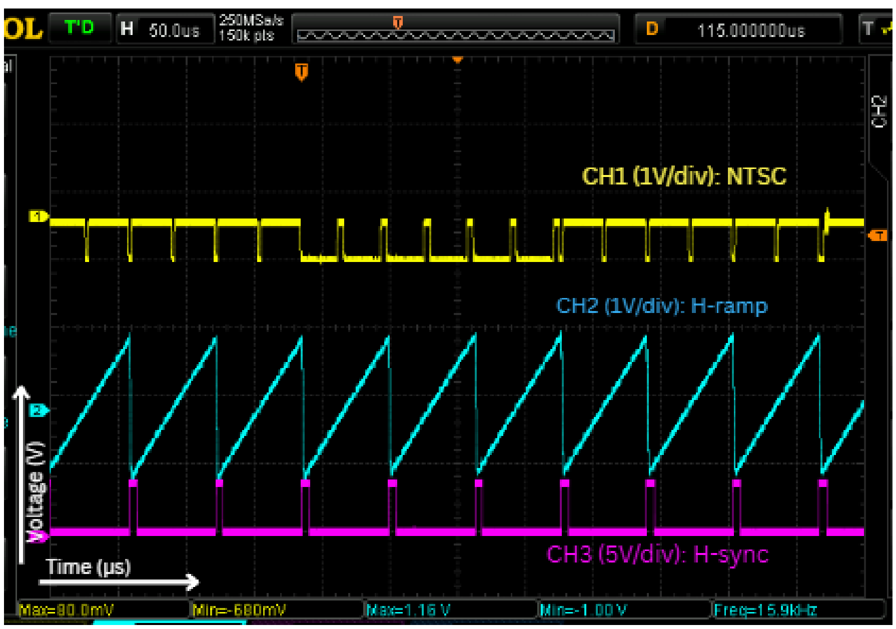

Figure 12: After H-sync Separator Change

Blanking Generator

Although a functional blanking generator was not successfully implemented in the final circuit. However, the intended design would generate a digital blanking flag that becomes active during both the horizontal and vertical retrace intervals. o the luminance-processing path. According to the project specification, this flag must remain active throughout the front porch, synchronization pulse, and back porch, rather than only during the synchronization pulse itself.

The horizontal and vertical sync signals could each trigger a monostable timing circuit. The horizontal monostable would generate a pulse wide enough to cover the complete horizontal retrace interval, while the vertical monostable would generate a longer pulse covering the complete vertical retrace interval. These two signals would then be combined using an OR function.

$$B_{blank} = B_H + B_V$$

B_H represents the horizontal blanking pulse and B_V represents the vertical blanking pulse. This would be so that blanking is active whenever either horizontal or vertical retrace is occurring. The resulting blanking flag would be sent to the luminance image-processing stage, where it would override the luminance signal and force the Z-axis output to the chosen black level. This is consistent with the specified system architecture, where the blanking generator feeds directly int

Luminance Separator

Since this subcircuit requires the removal of the synchronization information while only leaving the luminance information, a clipper circuit is used because sync pulses extend below the black level.

With a VD of 0.62 V, the clipping level is calculated as

$$V_{ref} = 12 \cdot \frac{2.2k}{39k+2.2k} = 0.641\, V$$

$$V_{clip} = V_{ref} - V_D = 0.641 - 0.62 = 0.02\, V$$

Thus the clipping level is 0.02 V

Since the majority of the luminance frequencies are on the lower end of the 4.2 MHz NTSC bandwidth, a bandwidth of 0 to 1.59 MHz was chosen. To do this, a passive low pass filter with a cutoff frequency of 1.59 MHz is used in order to isolate the luminance part from the chroma.

$$1.59\, MHz = \frac{1}{2\pi RC}.$$

Choosing R =100Ω

$$C = \frac{1}{2\pi(100)(1.59M)} = 1nF.$$

While selecting a 1.59 MHz cutoff frequency effectively attenuates the 3.58 MHz chroma color subcarrier, it restricts the luminance bandwidth well below its standard 4.2 MHz limit. This trade-off was intentionally made to prioritize stable sync separation and a subcarrier-free video baseline over high-frequency spatial details, resulting in a slightly softened image on the display.

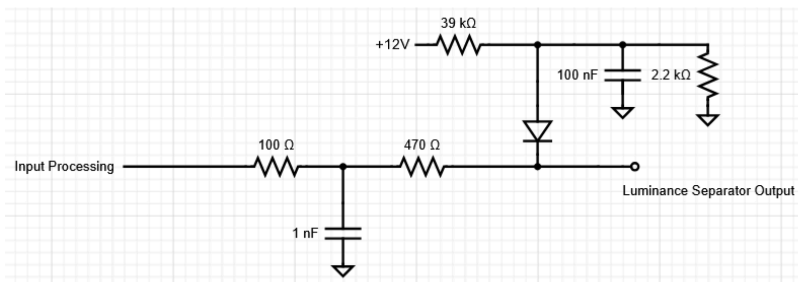

Figure 13: Luminance Separator Circuit

Luminance Image Processing

The luminance image processing stage should be able to adjust the contrast and brightness of the video with the use of trimpots, where contrast represents the gain and brightness represents the DC offset. Since a CRT cathode is being driven, the luminance signal also needs to be inverted because the CRT tube operates with an inverted brightness characteristic.

The LM6171 was used for the inverting amplifier due to its high bandwidth and slewrate. For the input network, a 10kΩ pot in series with a 1kΩ resistor was chosen for the inverting terminal along with a 100kΩ feedback resistor. These resistor values give a gain range of -9.1 V/V to -100 V/V, which is sufficient for a wide range of contrast adjustment.

$$A_{max} = -\frac{R_f}{R_1} = -\frac{100k}{1k} = -100\,V/V$$

$$A_{min} = -\frac{R_f}{R_1} = -\frac{100k}{10k+1k} = -9.1\,V/V$$

The brightness can be controlled by adjusting the DC offset of the output of the amplifier. To achieve this, the output is connected to the wiper of the trim pot, and the other 2 nodes are connected to 12V and ground through a 10k resistor. This creates a DC offset range of 6V to 12V.

$$V_{OS,\,min} = 12 \cdot \frac{10k}{10k} = 12\,V$$

$$V_{OS,\,min} = 12 \cdot \frac{10k}{10k+10k} = 6\,V$$

The output V_{Lumin} can be characterized by the equation.

$$V_{Lumin} = A * V_{Lumin\,Sep} + V_{OS}$$

Where A is the gain from the amplifier, $V_{Lumin\,Sep}$ is the incoming signal from the luminance separator, and V_{OS} is the offset.

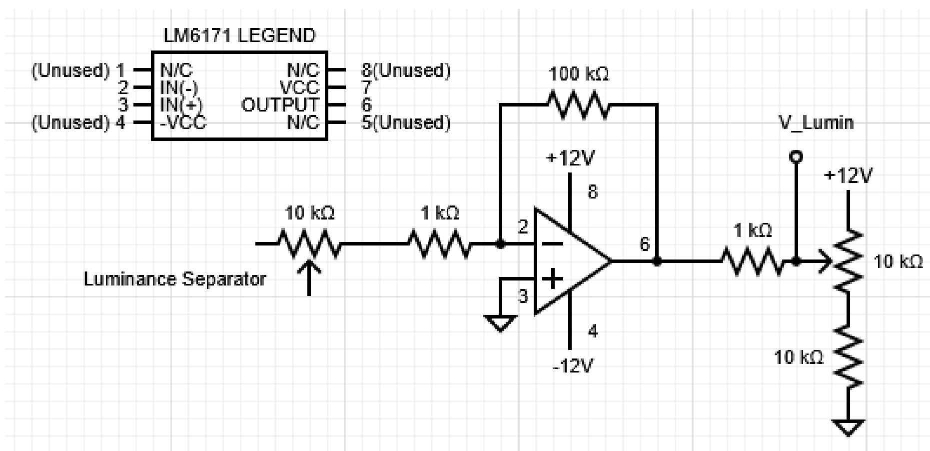

Figure 14: Luminance Image Processing Circuit

The Final System

Input Buffering, Clamping

Theory of Operation

The input buffering and clamping stage operates by first filtering the incoming video signal using a passive low-pass filter with a cutoff frequency of 1.59 MHz, attenuating the color burst signal while preserving the luminance signal. A diode clamper with a clamp voltage of 0.021 V is then used so that when the incoming NTSC signal falls below 0.021 V, the diode turns on, pulling the signal up to 0.021 V.

Measurement Procedure

1. Apply the artificial NTSC signal to the input clamping circuit and observe both the original input and processed output on the oscilloscope.
2. Measure the minimum voltage of the processed waveform and verify that the sync tips are clamped near the designed value of 0.021 V.
3. Compare the input and output waveforms to confirm that the 1.59 MHz low-pass filter attenuates the colour-burst/high-frequency components while preserving the main luminance waveform.
4. Verify that the processed artificial NTSC signal remains stable and sufficiently undistorted for use by the following composite sync separator and luminance stages.

Measurement Results

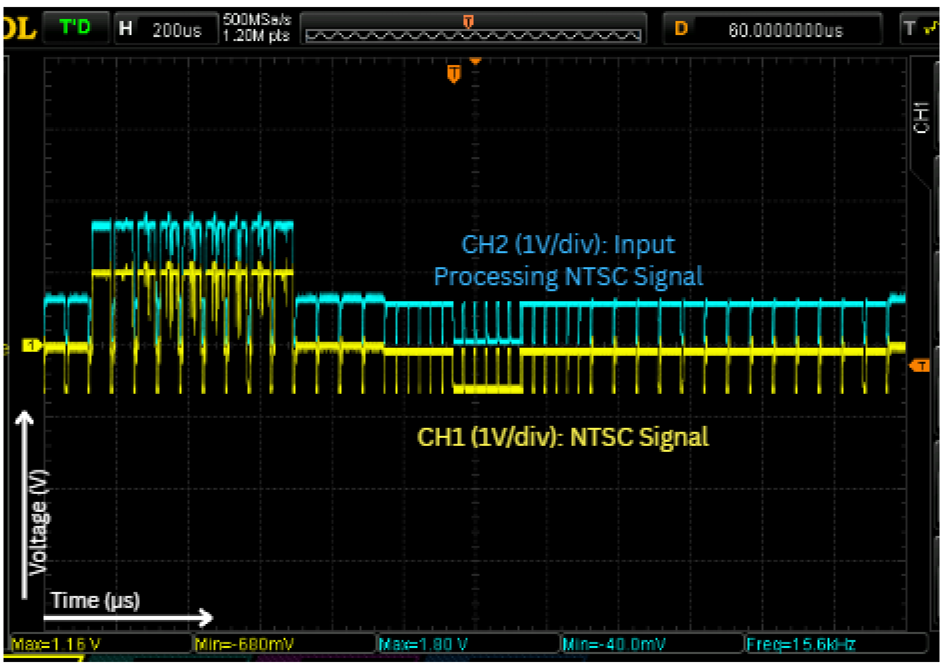

Figure 15: Input Processing NTSC and Artificial NTSC Oscilloscope Capture

Interpretation of Results

Disclaimer: This analysis applies exclusively to the circuit's performance with the synthetic NTSC signal; as noted previously in the design activity section for this subcircuit, the passive clamp experiences DC baseline drift when processing a live, dynamic NTSC video feed.

With both channels zero-positioned to the same baseline reference on the scope, Channel 2 (cyan) clearly shows the incoming NTSC video signal (Channel 1, yellow) shifted upward so that its lowest sync tips are clamped near 0V (ground). The clamped waveform perfectly preserves all structural components of the original signal,including active video, horizontal sync pulses, and the vertical sync interval, while successfully establishing the positive DC reference needed for downstream comparator processing.

Composite Sync Separator

Theory of Operation

The composite sync separator works by slicing the incoming video signal at a precise threshold to isolate the synchronization pulses from the active video content. With the reference voltage set to 176 mV at the inverting terminal (-), the comparator operates in an active-low configuration. The output is normally pulled up to 5 V, but whenever the non-inverting input (+) exceeds 176 mV, the comparator drives the output to 0 V (ground).

Measurement Procedure

1. Apply the NTSC signal and display the NTSC waveform and composite sync output on separate oscilloscope channels.
2. Measure the LM311 reference voltage and verify that the switching level is positioned between the sync-tip level and the blanking level of the clamped video signal.
3. Compare the two waveforms and confirm that each synchronization portion of the NTSC signal produces a corresponding composite-sync pulse, including the horizontal, equalization, and vertical sync pulses.
4. Vary the video content and observe the composite-sync output to ensure that the image information and colour burst do not produce unwanted pulses or missed sync transitions.

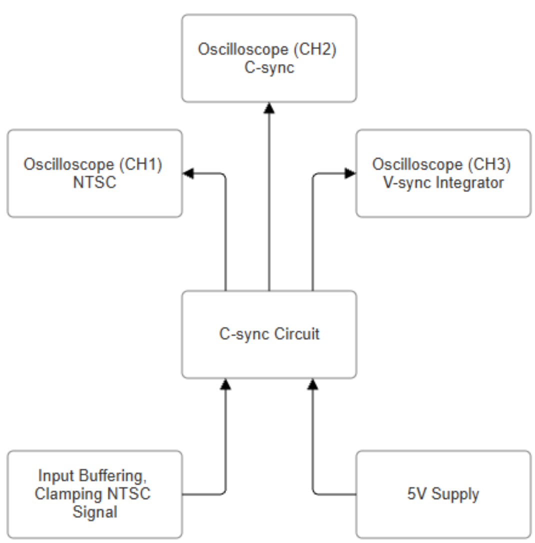

Figure 16: C-sync Block Diagram Connections

Measurement Results

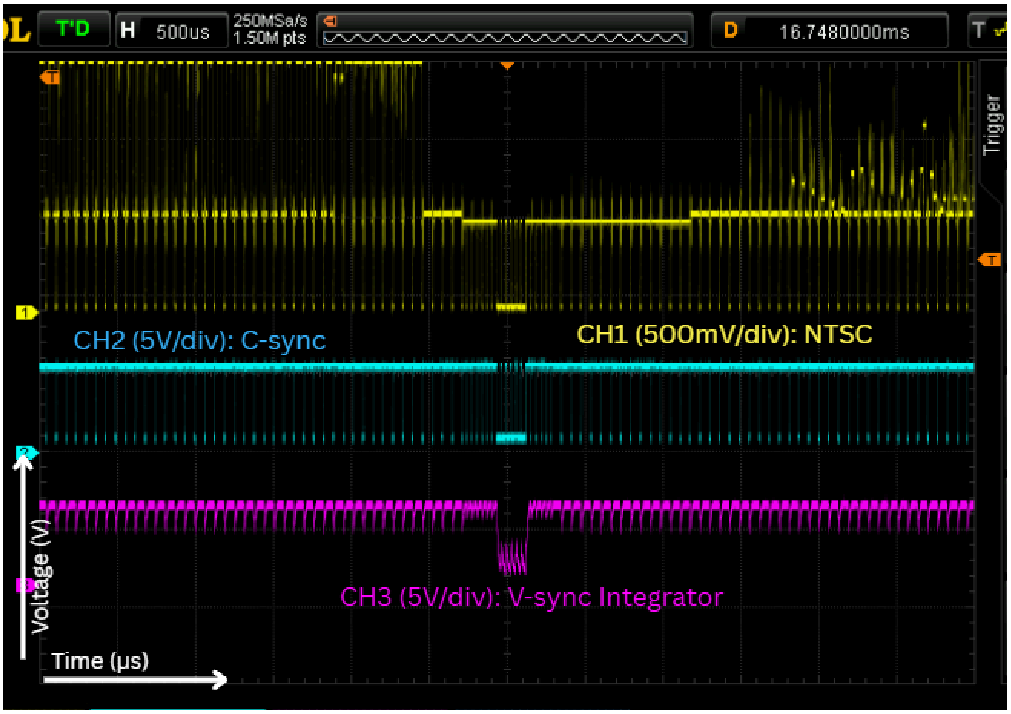

Figure 17: C-sync and V-sync Integrator Oscilloscope Capture

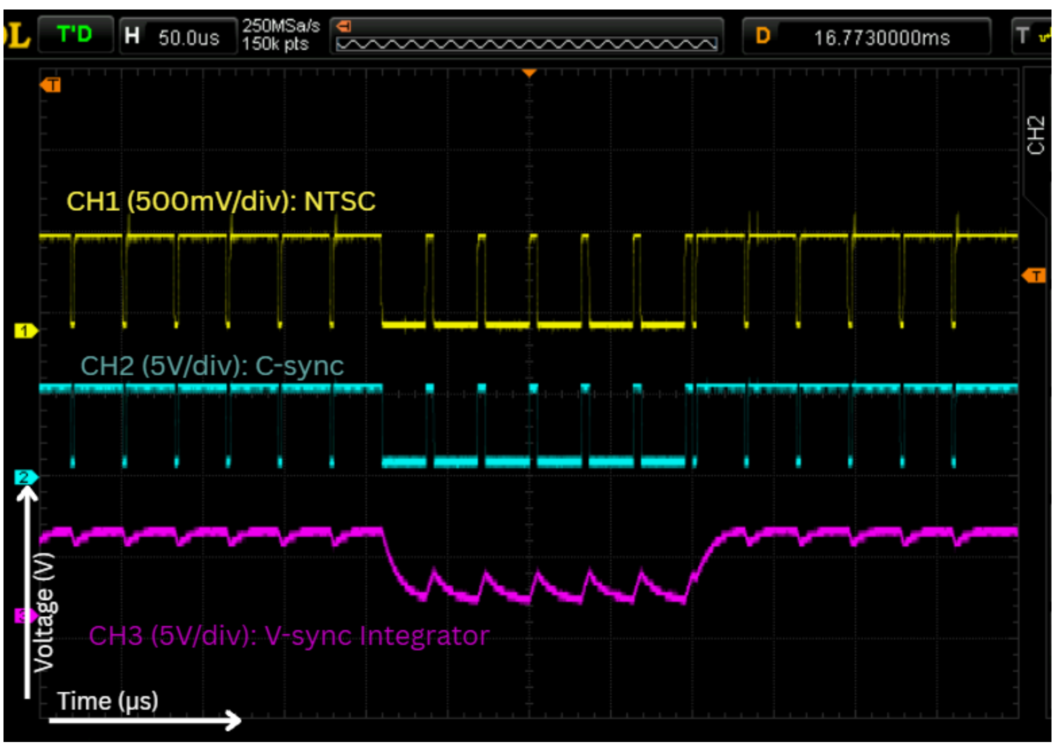

Figure 18: Zoomed C-sync and V-sync Integrator Oscilloscope Capture

Interpretation of Results

The composite sync separator demonstrates highly reliable pulse extraction, producing clean, sharp 5V logic transitions that reliably isolate synchronization pulses from the incoming NTSC video signal. The extracted output (Channel 2) successfully strips away all active luminance content while maintaining precise timing alignment with the input sync tips (Channel 1) during both horizontal scanning lines and the vertical retrace interval. Furthermore, the sharp pulse edges and well-defined high/low voltage levels show minimal jitter or threshold chatter, proving that the comparator stage processes equalization and vertical sync pulses accurately without false triggering or pulse degradation.

Horizontal Sync Separator

Theory of Operation

The first monostable multivibrator generates a 42.9 μs pulse to mask the double-frequency equalization pulses (31.75 μs period) during the vertical blanking interval, while leaving the standard 63.5 μs horizontal sync period undisturbed. The second monostable's TRIG pin (Node 8) is connected to both C-sync and the emitter of an NPN transistor, creating a hardware AND logic condition for triggering. The first condition requires the output of the first monostable to be LOW; when its output is HIGH, it turns on the NPN transistor (configured as an emitter follower with its collector at +5V), forcing Node 8 up to approximately 4.3V and overriding any incoming trigger edges. The second condition requires a falling edge from C-sync to pull Node 8 below ⅓VCC (1.67V). When both conditions are met, Timer 1 is LOW and a C-sync falling edge arrives, the second monostable triggers, generating a pulse width of 5.17 μs.

Measurement Procedure

1. Probe C-Sync and the output first monostable of H-Sync to see that all the double frequencies have been masked.
2. Probe the C-sync signal alongside the outputs of the first and second H-sync monostable stages to verify that the second monostable fires exclusively when the first monostable output is LOW and during a horizontal sync tip.
3. Confirm the H-sync frequency is 15.734 kHz by measuring the period during a normal line.

4. Use the oscilloscope to determine the H-Sync pulse duration and verify that it is close to 4 µs.
5. Observe the signal during the vertical retrace period and check that the H-Sync pulses remain continuous without any skipped or additional pulses through the equalization and serration intervals.
6. Compare the timing of the Composite Sync transition with the corresponding H-Sync transition and measure the delay between them. Verify that this delay is negligible compared with the duration of one horizontal line.

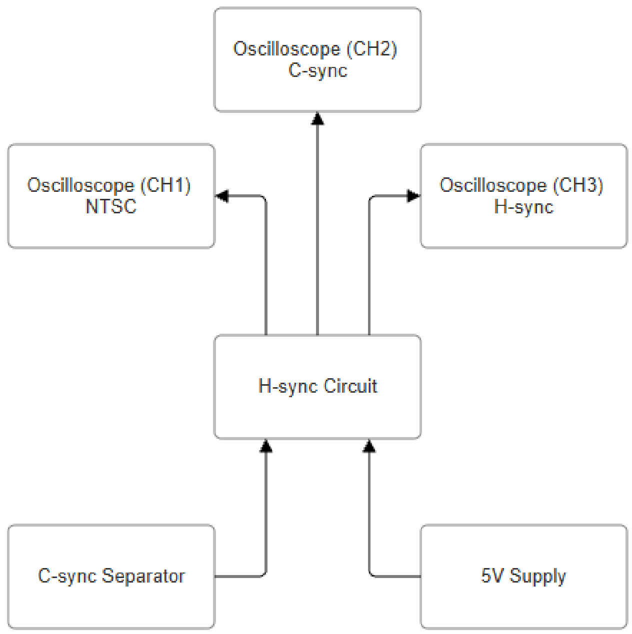

Figure 19: H-sync Block Diagram Connections

Measurement Results

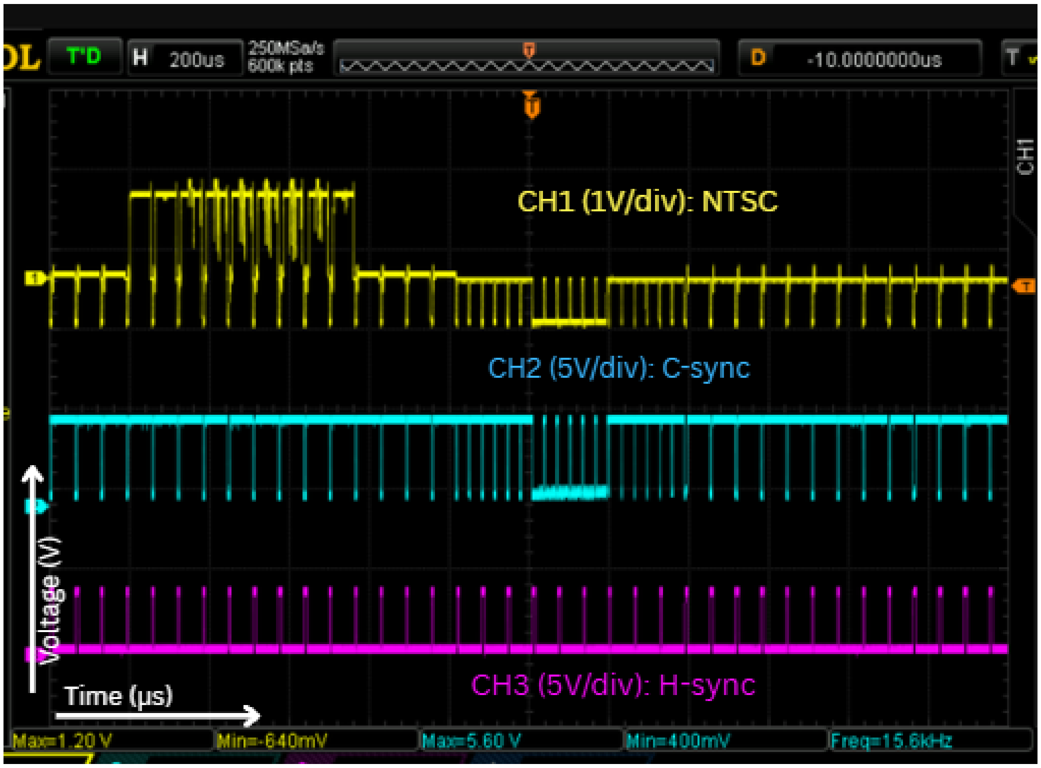

Figure 20: H-sync and C-sync Oscilloscope Capture

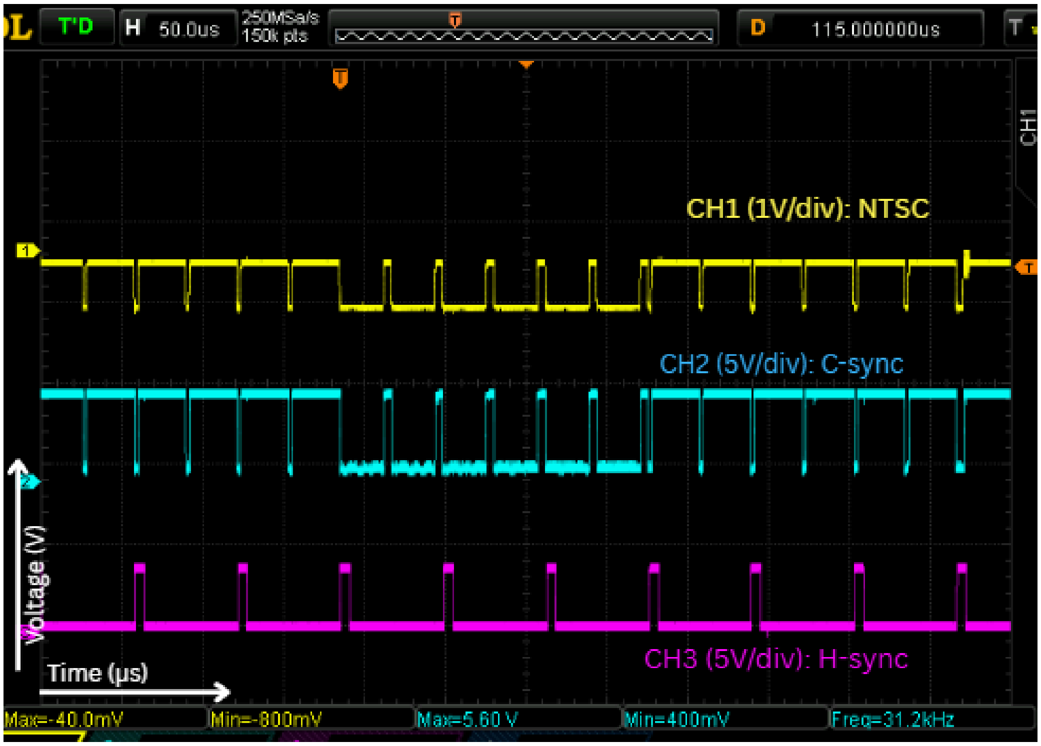

Figure 21: Zoomed H-sync and C-sync Oscilloscope Capture

Interpretation of Results

The output on Channel 3 (magenta) successfully isolates a continuous, uniform stream of horizontal sync pulses from the composite signal, maintaining a consistent frequency (15.734 kHz) even through the vertical retrace interval. A slight propagation delay is visible between the falling edge of the input composite sync signal (Channel 2, cyan) and the rising edge of the separated horizontal sync pulse (Channel 3, magenta), this delay is acceptable as it remains well within the timing constraints of the overall system design.

Vertical Sync Separator

Theory of Operation

The V-sync separator works by filtering out high-frequency horizontal sync pulses and extracting only the longer vertical synchronization interval. The C-sync signal passes through a low-pass filter with a cutoff frequency of 1591 Hz, attenuation that allows only the broader V-sync pulse energy to remain. The filtered output is then fed into a 555 timer stage to add hysteresis, preventing output chattering or false triggers caused by residual high-frequency noise.

Measurement Procedure

1. Display the composite sync signal and the V-sync output from the NE555 on separate oscilloscope channels over several video fields.

2. Measure the repetition rate of the V-sync output and verify that it occurs at approximately 60 Hz.
3. Check that the circuit produces a single V-sync pulse for each field and does not respond to the normal horizontal sync pulses.
4. Measure the delay between the vertical-sync portion of the composite sync waveform and the generated V-sync pulse, and verify that the delay remains within the specified 25% of one line period.
5. Change the video content and confirm that the vertical sync output remains stable without extra or missing pulses.

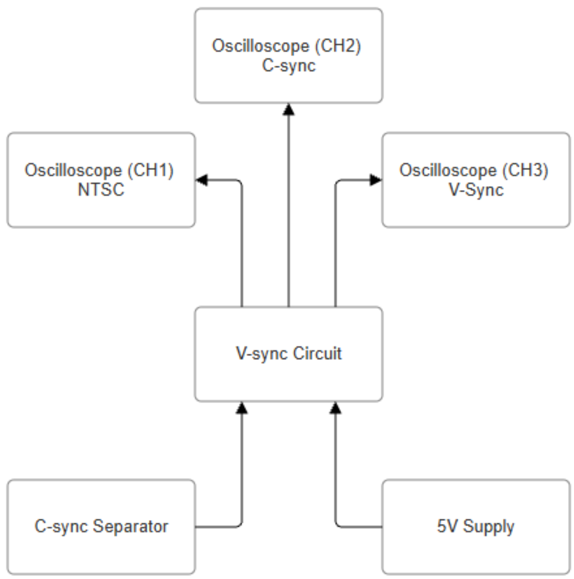

Figure 22: V-sync Block Diagram Connections

Measurement Results

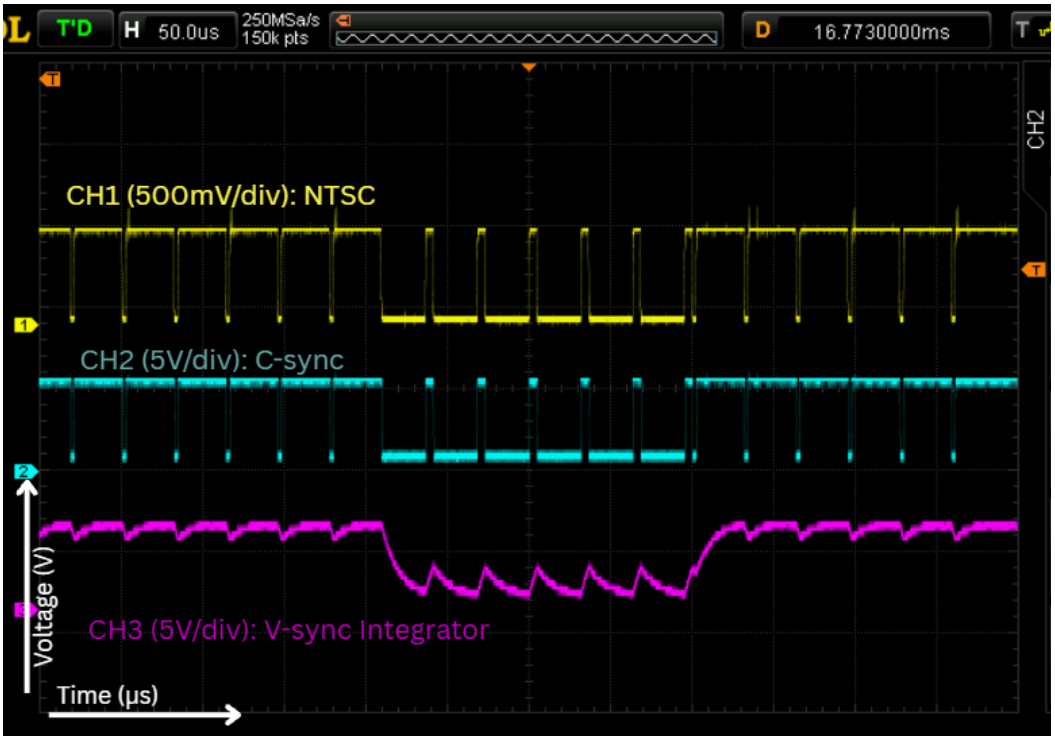

Figure 23: C-sync and V-sync Integrator Oscilloscope Captures

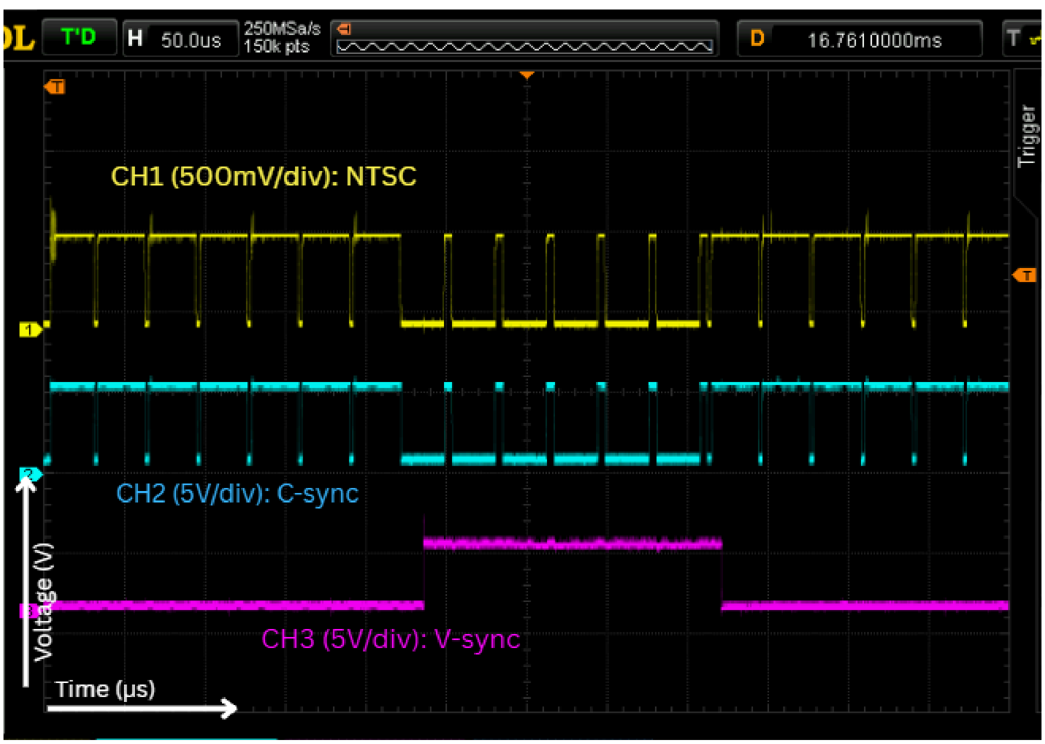

Figure 24: V-sync and C-sync Oscilloscope Capture

Interpretation of Results

In the first image, the low-pass filter successfully integrates the longer vertical sync pulses into a distinct downward voltage ramp while leaving the narrow horizontal pulses as minor high-frequency ripples. In the second image, the 555 timer's hysteresis successfully ignores those high-frequency ripples and cleanly triggers on the integrated vertical pulse, generating a crisp, square vertical sync pulse for the duration of the retrace interval.

Horizontal and Vertical Sweep Generators

Theory of Operation

The horizontal and vertical ramp generators work by using the sync separator pulses to periodically discharge a capacitor that is continuously charged by a constant current source, producing a linear voltage ramp. When the subcircuit receives a LOW output(0V) from the H-sync or V-sync separator the DC offset will pull the voltage to 7.2V, but since threshold level is at ⅔ V_{CC} (8V) the top internal comparator will send a LOW signal R=0 keeping its previous state. This allows the capacitor to charge up, even though the capacitor is connected to DISCHARGE and TRIG pins, the internal transistor never turns on unless the input from the H-sync or V-sync separator surpasses the threshold level. When the subcircuit receives a HIGH output (5V) from the H-sync or V-sync separator the DC offset will pull the voltage to 12.2V surpassing the threshold level and turning on the transistor and dissipating the voltage in the capacitor until the capacitor reaches below ⅓ V_{CC} (4V) causing the internal bottom comparator to turn off the transistor then causing the capacitor to recharge, repeating the cycle.

Measurement Procedure

1. Apply the NTSC video signal and observe the H-ramp and V-ramp outputs on the oscilloscope.
2. Compare each ramp with its corresponding sync signal to verify that the ramp resets at the correct time.
3. Measure the ramp frequency, peak-to-peak amplitude, and reset time for both the horizontal and vertical ramps.
4. Check ramp linearity by measuring several points along the active portion of each ramp.
5. Connect the H-ramp and V-ramp to the oscilloscope X and Y inputs and verify that a stable raster is produced.

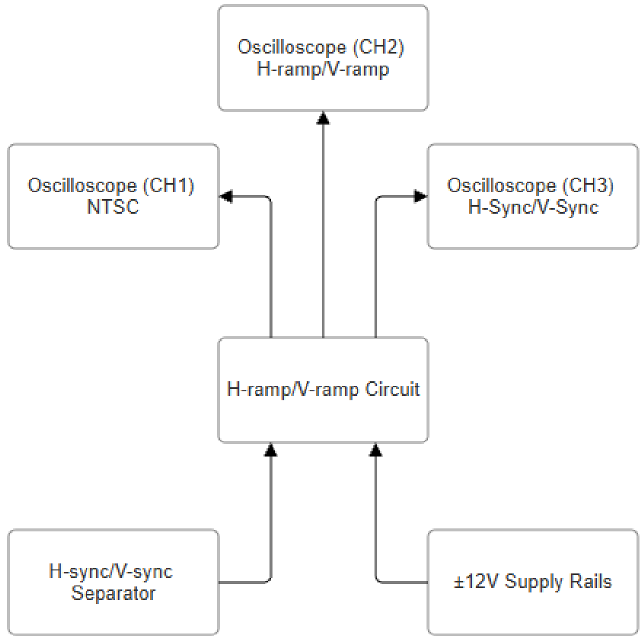

Figure 25: H-ramp and V-ramp Block Diagram Connections

Measurement Results

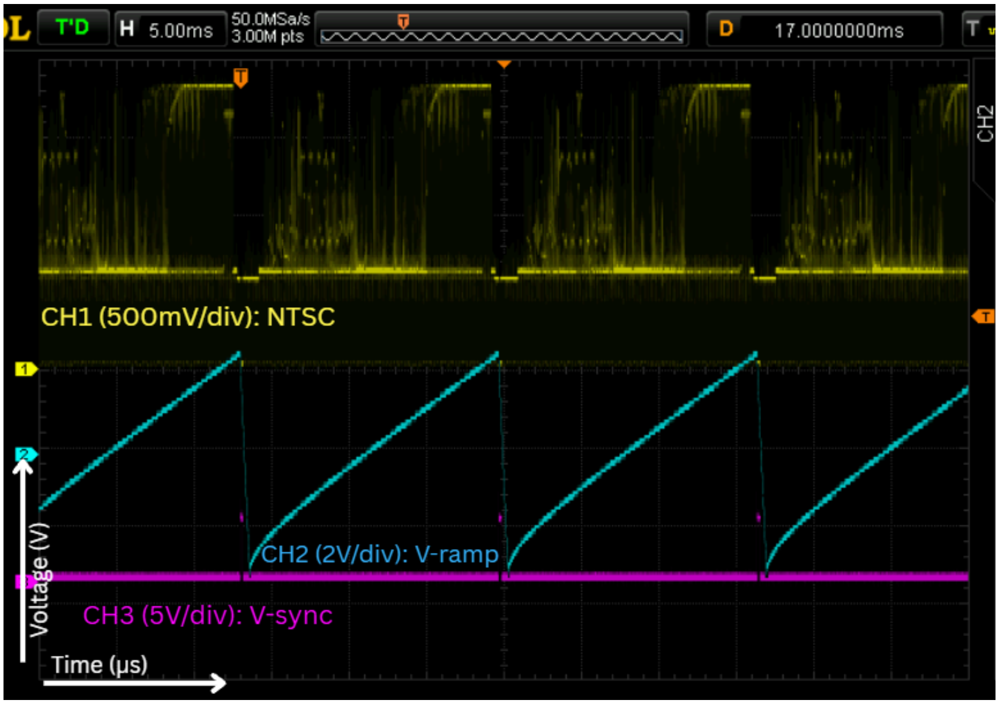

Figure 26: V-sync and V-ramp Oscilloscope Capture

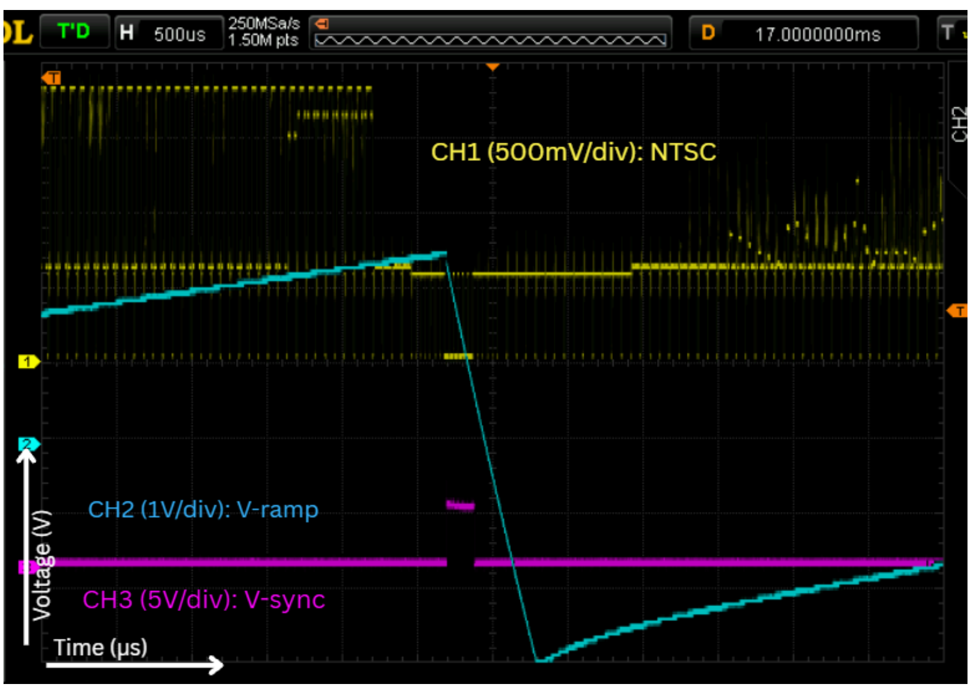

Figure 27: Zoomed V-sync and V-ramp Oscilloscope Capture

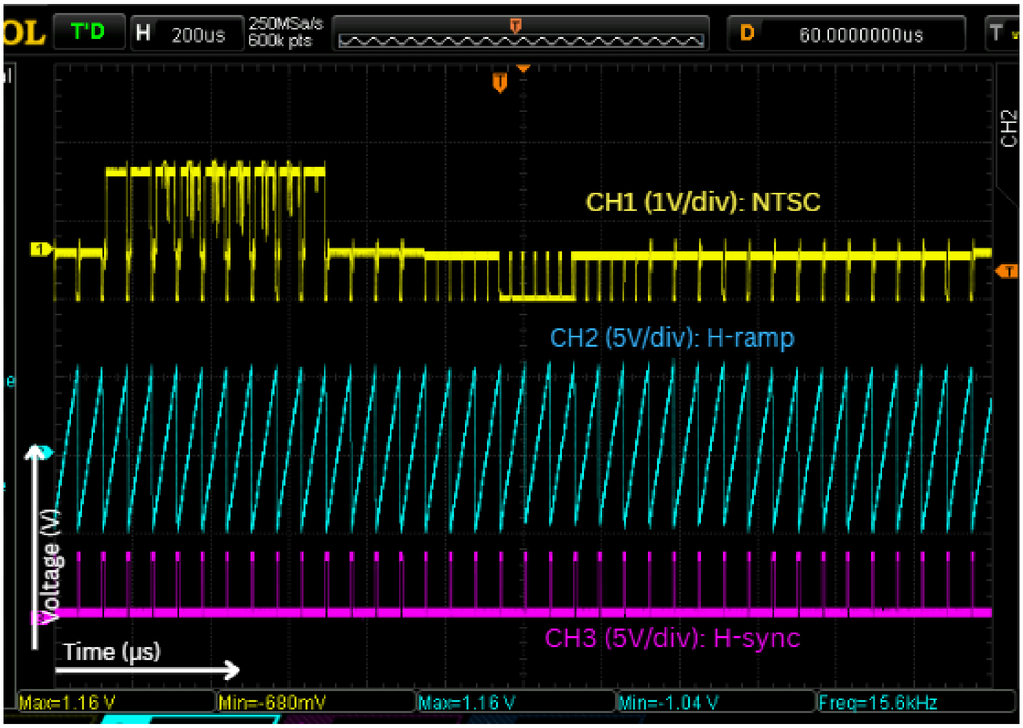

Figure 28: H-ramp and H-sync Oscilloscope Capture

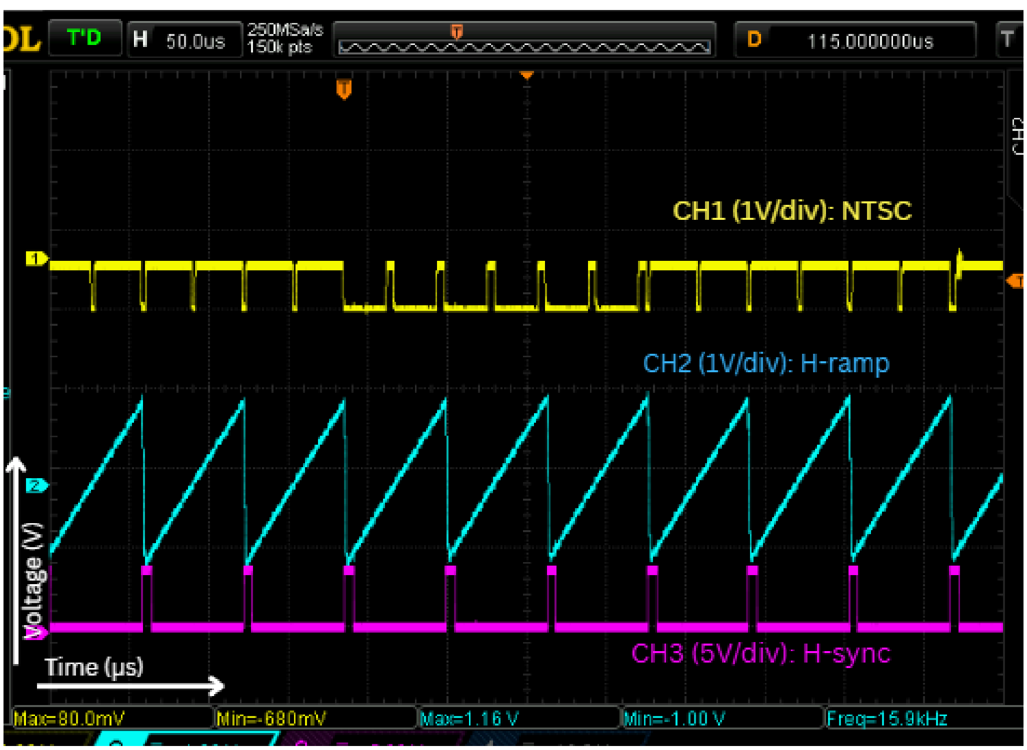

Figure 29: Zoomed H-ramp and H-sync Oscilloscope Capture

Interpretation of Results
The sweep generation subcircuits demonstrate clean linear ramp outputs for both horizontal and vertical deflection. The horizontal ramp generator outputs a ~63.5 μs waveform at the line rate and the vertical ramp generator operating at a ~16.67 ms period (60 Hz field rate), each discharging smoothly upon receiving its respective extracted sync pulse (CH3) with minimal delay to ensure stable frame synchronization.

Blanking Generator

Disclaimer: The blanking generator circuit was not implemented in the final hardware system. Consequently, measurement results and result interpretations are omitted. The procedure below outlines the intended verification steps for the subcircuit.

Theory of Operation
The blanking generator uses the horizontal and vertical synchronization signals to generate a blanking flag during the retrace portions of the NTSC signal. The horizontal blanking signal remains active throughout the horizontal front porch, sync tip, and back porch, while the vertical blanking signal covers the corresponding vertical retrace interval. These signals are combined using an OR function so that the blanking flag is active whenever either horizontal or vertical retrace occurs. The blanking flag is then provided to the luminance image-processing stage, where it forces the Z-axis signal to the specified black level during retrace and prevents unwanted retrace lines from appearing on the CRT.

Measurement Procedure
The blanking generator was not successfully implemented in the final system. Therefore, the following procedure describes how the subcircuit would have been tested if it were functional.

1. Display the H-sync, V-sync, and blanking output signals on the oscilloscope.
2. Verify that the blanking flag is active throughout both the horizontal and vertical retrace intervals.
3. Connect the blanking output to the luminance-processing stage and verify that the Z-axis output is forced to the specified black level during retrace.

Luminance Separator

Theory of Operation
The luminance separator isolates the active video intensity information from the composite NTSC signal by stripping away synchronization pulses and color subcarrier interference by using a low pass filter with a cutoff frequency at 1.59 MHz to attenuate the 3.58 MHz chrominance

subcarrier and a clipping stage to truncate all negative-going sync tips below the black-level threshold, leaving only active luminance content.

Measurement Procedure

1. Apply the NTSC video signal and observe the clamped NTSC waveform together with the luminance separator output on the oscilloscope.
2. Verify that the sync portions of the waveform are removed or clipped while the active luminance information remains visible.
3. Change the image brightness and check that the luminance waveform still varies normally without the sync pulses reappearing at the output.
4. Measure the response of the luminance path using a sharp video transition or suitable test signal to estimate its bandwidth and confirm that it is sufficient to preserve the image detail.

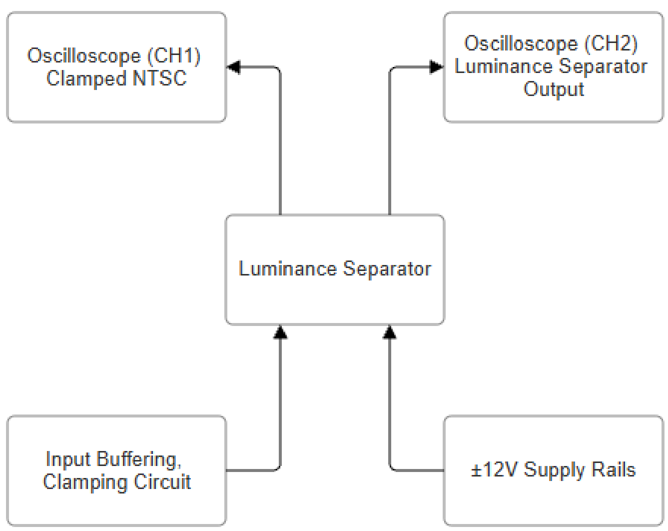

Figure 30: Luminance Separator Block Diagram Connections

Measurement Results

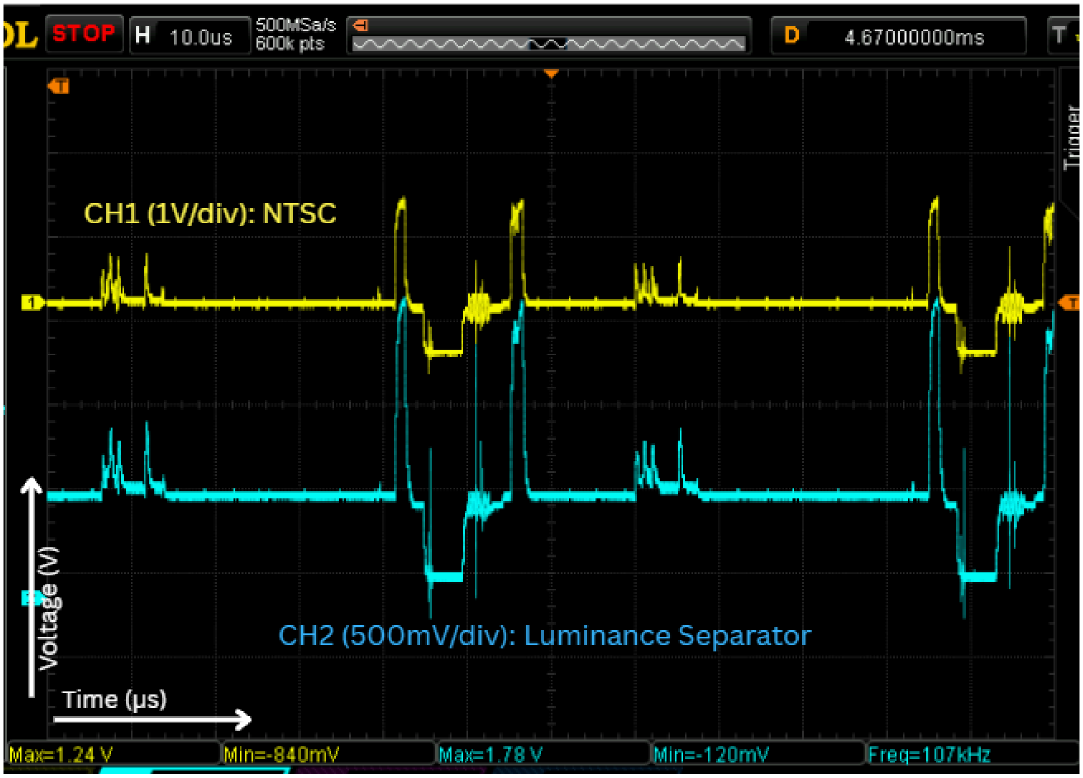

Figure 31: Luminance Separator Oscilloscope Capture

Interpretation of Results

The oscilloscope capture shows that the luminance separator is operating as intended by removing most of the negative-going synchronization content from the original NTSC signal while preserving the changing voltage levels that represent the video luminance information. Comparing CH1 and CH2, the large sync pulses present in the NTSC waveform are clipped to a nearly constant level at the separator output, while the smaller variations associated with image brightness remain visible. Some residual spikes and high-frequency components are still present due to non-ideal clipping and filtering, but the resulting waveform is suitable for use by the following brightness and contrast stages.

Luminance Image Processing

Theory of Operation

The luminance image processing works by scaling active video signal amplitude and setting the black-level offset using an op-amp configured as an inverting amplifier. The input video signal is fed through a 1kΩ resistor in series with a 10kΩ potentiometer which controls contrast into the inverting terminal, where a 100kΩ feedback resistor establishes an adjustable inverting voltage gain range of -9.1V to -100V. A second 10kΩ potentiometer which controls brightness is connected to the output of the amplifier and biased between 12V and ground through a 10kΩ. Creating a DC offset range of 6V to 12V. The contrast and brightness potentiometers operate independently because the contrast pot varies the input resistance to scale only the AC signal gain, while the brightness pot injects a variable DC bias into the summing node to shift only the DC baseline voltage.

Measurement Procedure

1. Apply an NTSC test image with several brightness levels and observe the processed luminance output at the oscilloscope Z-input.
2. Adjust the contrast control through its range and measure the change in luminance amplitude, checking that the blanking/black level remains approximately unchanged.
3. Adjust the brightness control and observe the resulting shift in output level while checking that the luminance amplitude is not significantly altered.
4. Measure several input and output luminance levels at selected control settings to verify that the processing stage remains approximately linear and to identify any clipping at the limits.
5. Connect the final luminance signal to the oscilloscope Z-axis with the X- and Y-ramps active, then verify that brightness and contrast controls produce the expected visual changes in the displayed image.
6. Observe the Z-axis signal during horizontal and vertical retrace and confirm that the blanking signal forces the output to the specified black level regardless of the brightness and contrast settings.

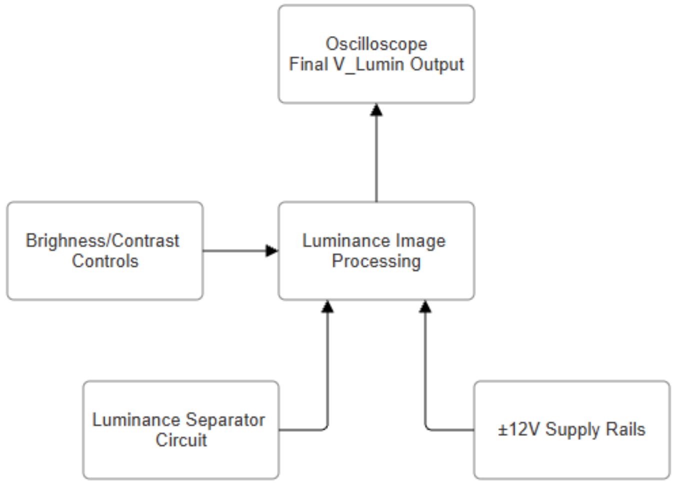

Figure 32: Luminance Image Processing Block Diagram Connections

Measurement Results

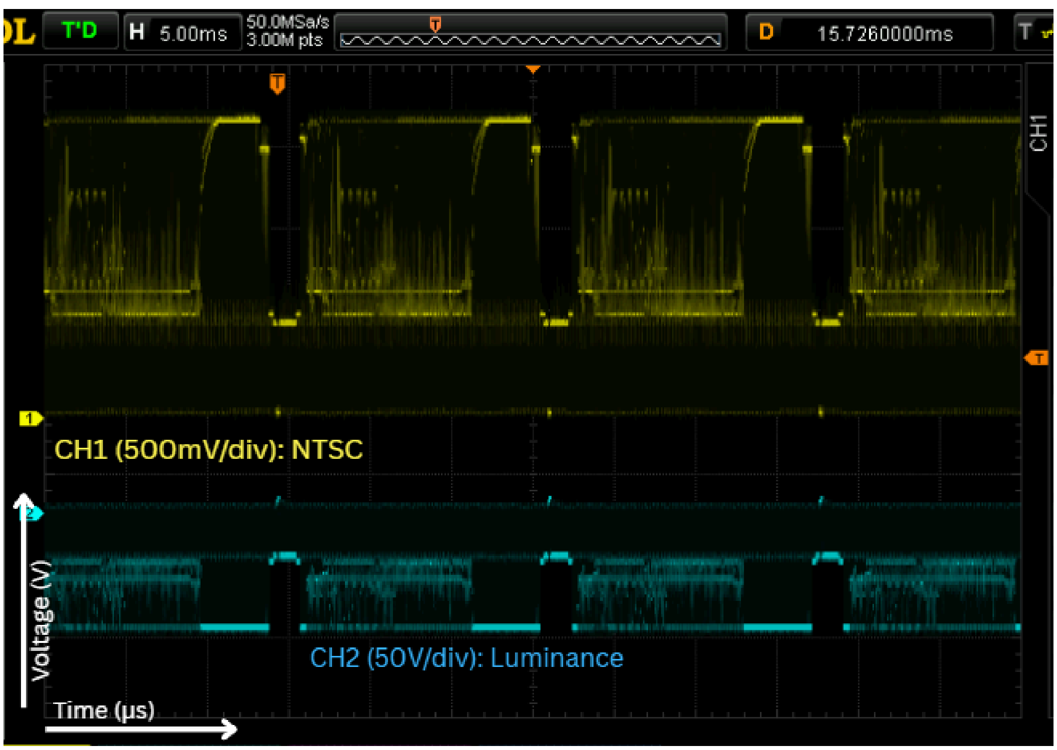

Figure 33: Luminance Image Processing Oscilloscope Capture

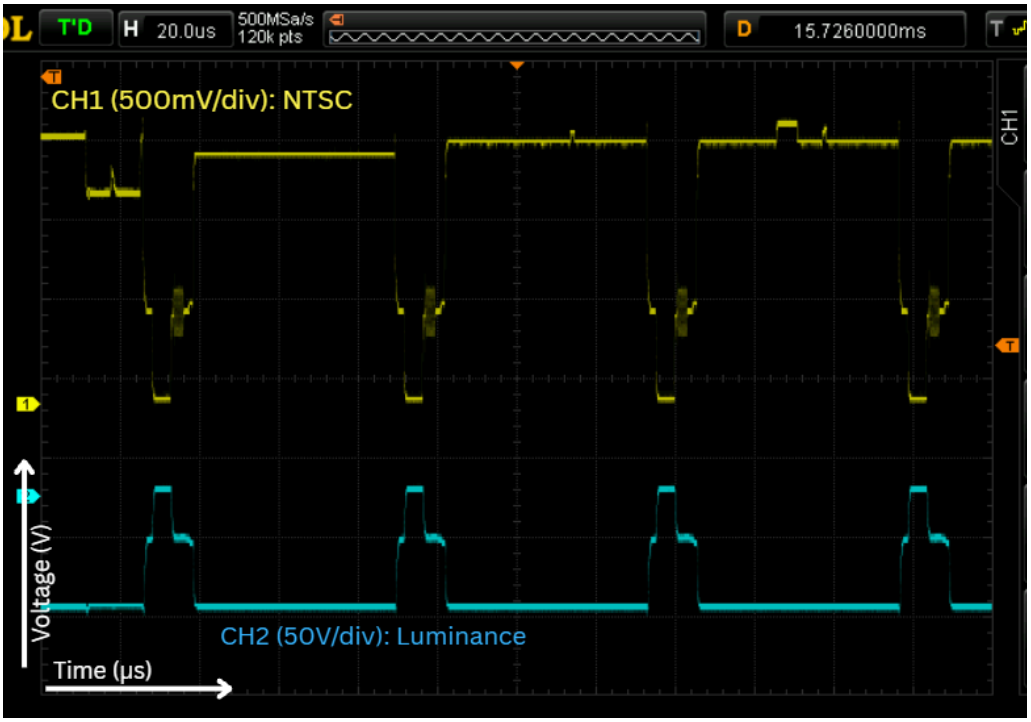

Figure 34: Zoomed Luminance Image Processing Oscilloscope Capture

Interpretation of Results

The oscilloscope captures show that the luminance image processing stage successfully conditions the separated luminance signal into a form suitable for driving the oscilloscope Z-axis. Compared with the original NTSC waveform, the processed CH2 signal has a shifted DC level and a larger, more clearly defined amplitude corresponding to the video brightness information. The zoomed capture also shows that the processed luminance follows the active-video portions of the NTSC signal while remaining relatively constant during the non-image intervals. Overall, the stage provides the expected brightness and contrast adjustment while preserving the main luminance variations needed to reproduce the image

Conclusion

In this project, an analog video decoding system was designed, built, and tested to process a standard NTSC composite video signal and output the corresponding X-, Y-, and Z-axis signals required to display video raster images on an oscilloscope CRT. The system integrated input clamping and filtering, composite sync extraction, horizontal and vertical sync separation, sweep ramp generation, luminance separation, and inverting luminance image processing.

Key design objectives were successfully realized: a passive diode clamp restored a stable positive DC baseline near 0.021 V, while a 1.59 MHz low-pass filter with a 1 MΩ discharge path attenuated the 3.58 MHz color subcarrier to ensure clean reference levels and prevent node floating. For timing separation, an LM311 comparator extracted composite sync tips at a 176 mV threshold, while interlocked dual-stage 555 monostable multivibrators masked double-frequency equalization pulses (31.468 kHz) to output a continuous 15.734 kHz horizontal sync stream.

An RC integrator (15.9 kHz) combined with a 555 hysteresis stage cleanly extracted 60 Hz vertical sync pulses, while Wilson current sources driving 555 timers produced linear horizontal (~63.5 µs) and vertical (~16.67 ms) sweep ramps with fast retrace times. Additionally, clipping below the black level and amplifying with an LM6171 high-speed op-amp enabled independent gain and DC offset adjustments for CRT Z-axis contrast and brightness control.

While core raster scanning and luminance processing were fully functional, hardware trade-offs remained, including the lack of a fully integrated dedicated blanking generator relying instead on standard luminance level control and minor DC baseline drift under dynamic live NTSC video due to changing Average Picture Levels (APL). Overall, the design successfully validated the principles of analog video timing separation, sweep synchronization, and high-frequency active signal processing, with future improvements pointing toward an active DC restoration clamp and dedicated logic gates for complete retrace blanking.

Appendix

Circuit Schematics

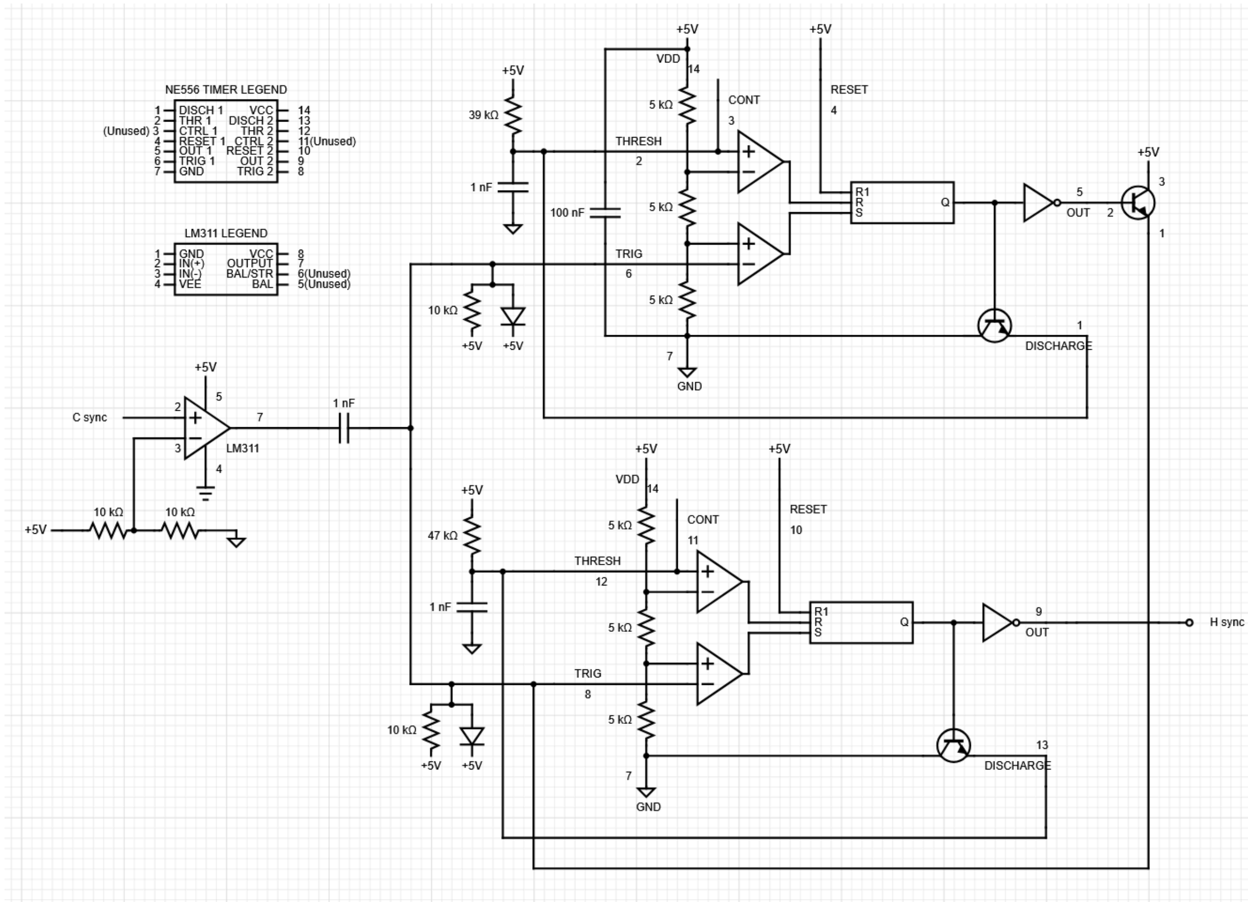

Figure 35: H-sync Circuit Diagram (3rd iteration)

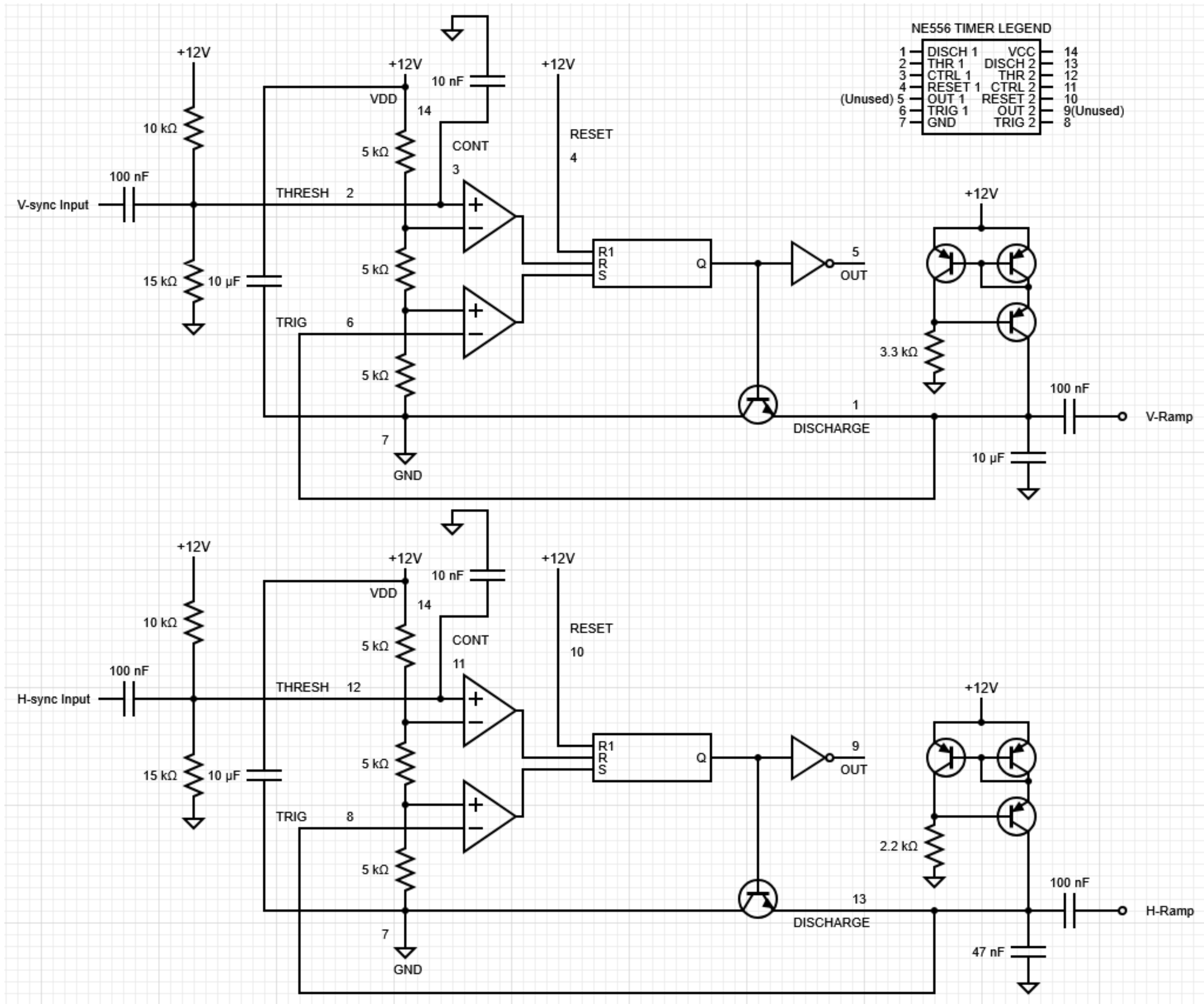

Figure 36: H-ramp and V-ramp (2nd Iteration)

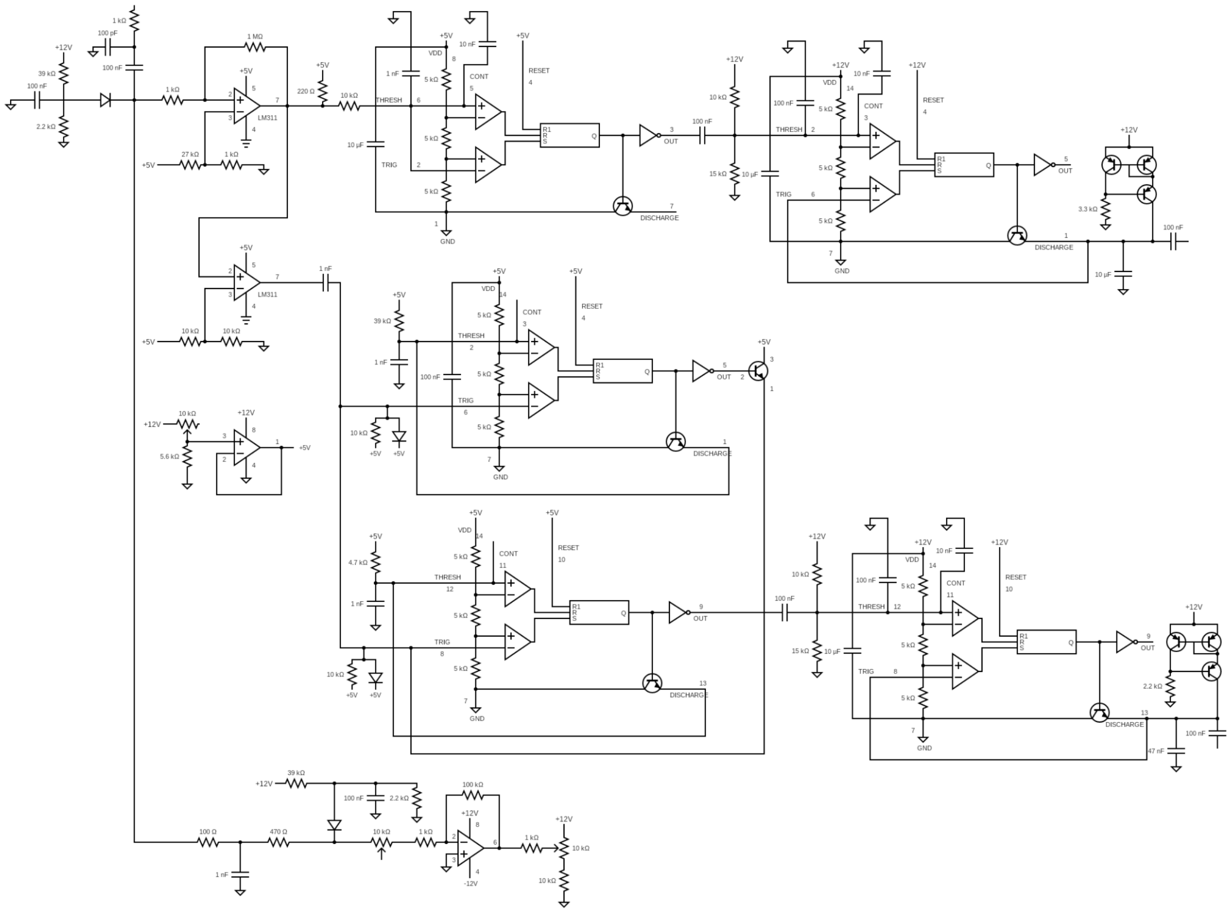

Figure 37: Full Circuit Schematic

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